
Structural Engineering- An Overview

Barzin Mobasher, PhD. PE., FACI, F-RILEM
Narayanan Neithalath, Ph.D., FACI

School of Sustainable Engineering and the Built Environment
Arizona State University
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Structural Engineering: Temporal, Spatial, and Scientific Span



Disciplines

- Materials Science
- Engineering
- Chemistry
- Mechanics
- Computational Techniques
- Manufacturing products and systems
- Sustainable development
- Technical & non-technical labor pool

Time Scale

hydration Early age Service life Long term Performance

nanometers to kilometers
(1×10^{-8} to 1×10^3 meters)

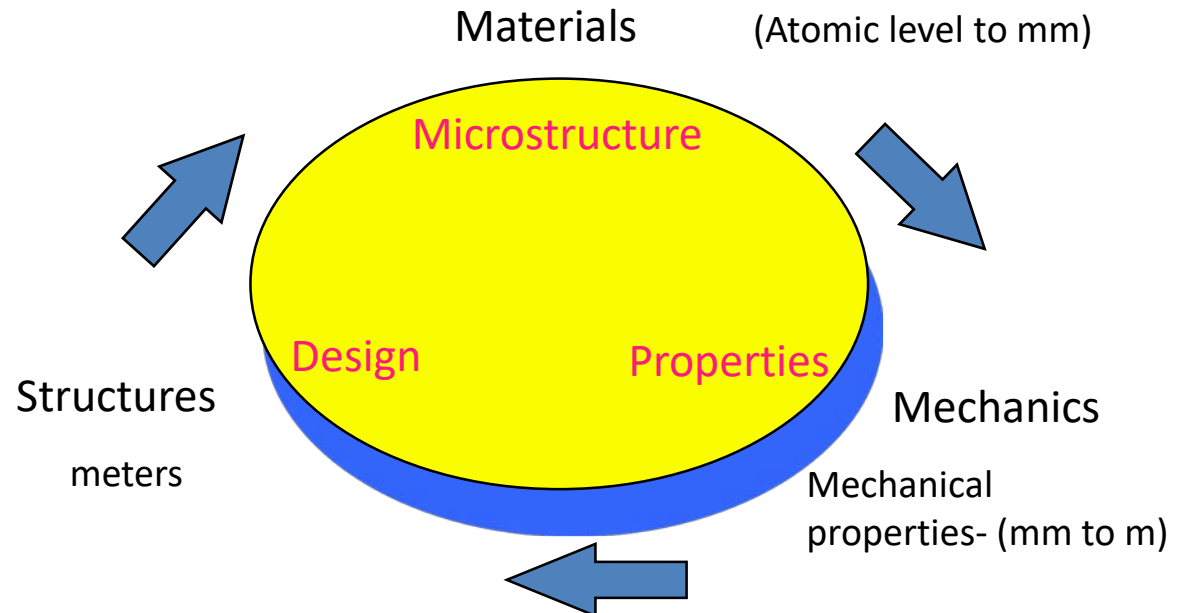
Length Scale

Seconds to Centuries
(1 to 3×10^{10} Sec.)



Structures?

- Low-rise and high rise buildings
- Infrastructure, Parking garages, facilities
- water tanks, hydraulic channels
- Dams, domes, bridges, transmission towers,
- nuclear containment vessels
- aircrafts and space structures, packaging and IC chips
- automobile components, cage, chassis, bumpers
- biomechanics, surgical implants



Materials-Mechanics-Structures Interaction

Societal Challenges in Structural Engineering

- Control of the Environment
- Safe, secure, and comfortable shelter
- Mitigation of Natural Disasters
 - Earthquakes, Hurricanes & Tsunamis, Fire, and Flood.
- Functioning Infrastructure for Economic Development

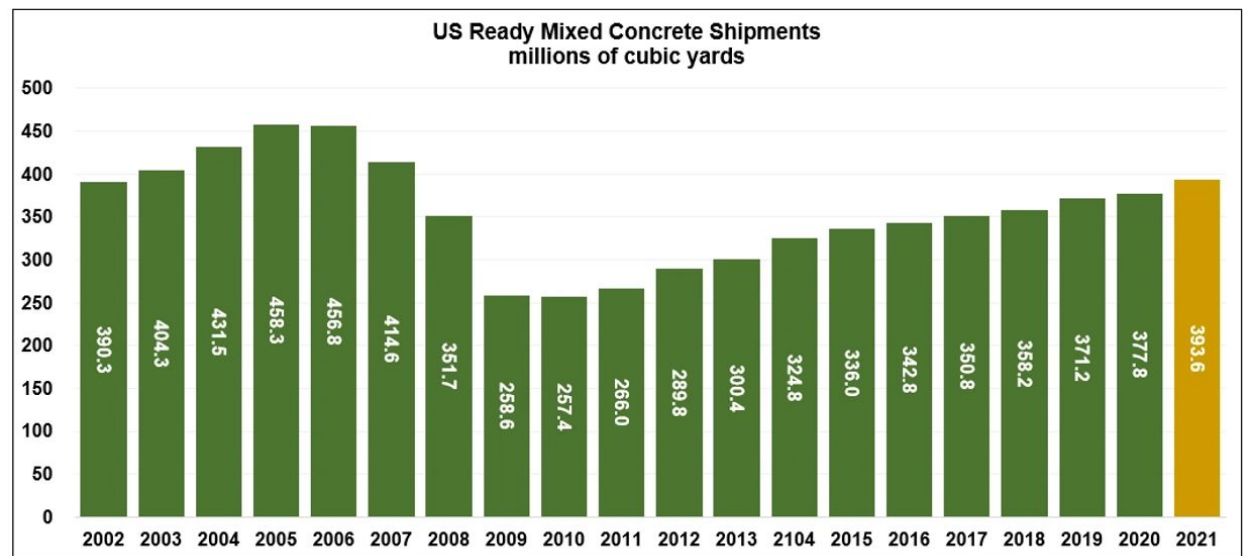


Evolution of Structural Engineering

- Societal Challenges
 - What are the challenges that we face in the next decade?
 - Owner, high initial investment, 50+ year service life, Conservatism in design
 - Sustainability
 - Reuse and recycle, new construction systems
 - Design for serviceability
 - Quality control, materials and design
 - Traditional & new systems, 30+ years needed for market penetration
 - Integrated Design and Analysis Systems
 - Development of design guides using computational tools
 - Modeling, simulation, life cycle assessment tools, acceptability, documentation and durability of new materials
 - Repair and Retrofit
 - Earthquake Mitigation
 - Ductility Based Design
 - Durability
- Damage mitigation

Sustainability of Construction Materials

- The Bipartisan Infrastructure Law (Infrastructure Investment and Jobs Act), a once-in-a-generation investment in our nation's infrastructure and competitiveness. (Nov.2021)
- Concrete is the 2nd most consumed material. One ton per person per year
- How do we address sustainability?
 - Efficient use of resources, less CO₂ production, (97% of energy use of building is in service operation and only 3% in initial costs)
 - Novel construction techniques and materials, Extended Service Life.
 - Development of Design and Analysis tools.



Source: NRMCA utilizing US Geological Survey data with Concrete Financial Insights estimate for 2021 based on trend

Population Growth & Wildfires





Hurricane Katrina, August, 2005, Long Beach, Mississippi- Reinforced Concrete House

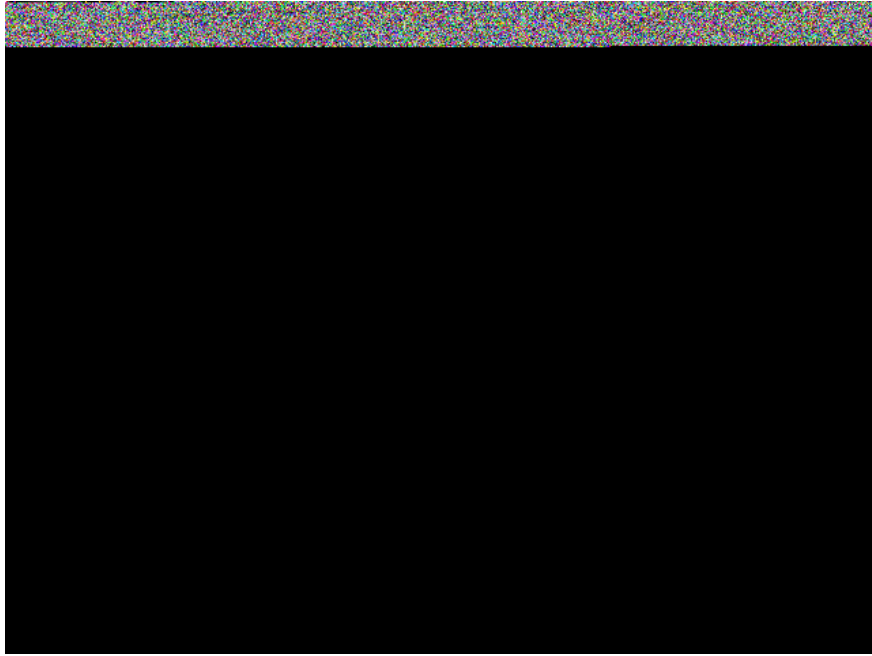


Photo Credit: FEMA/Mark Wolfe

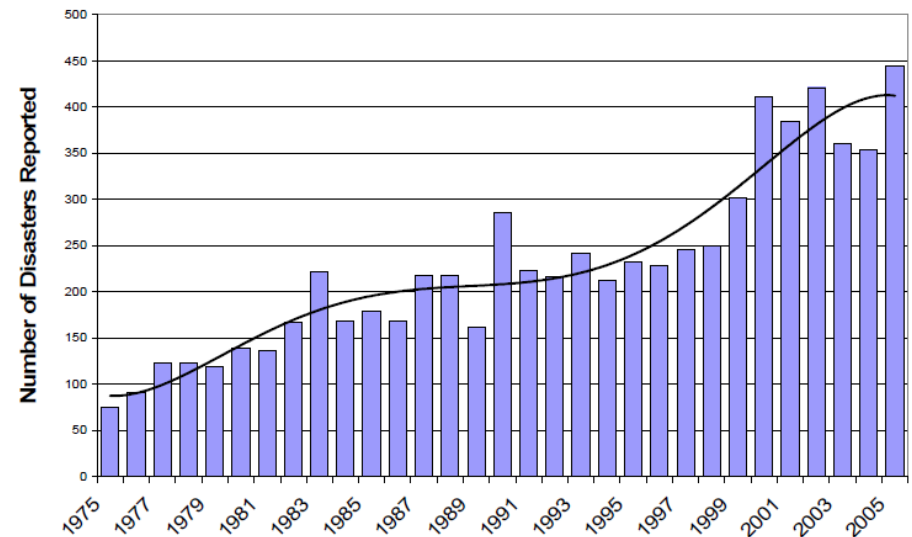
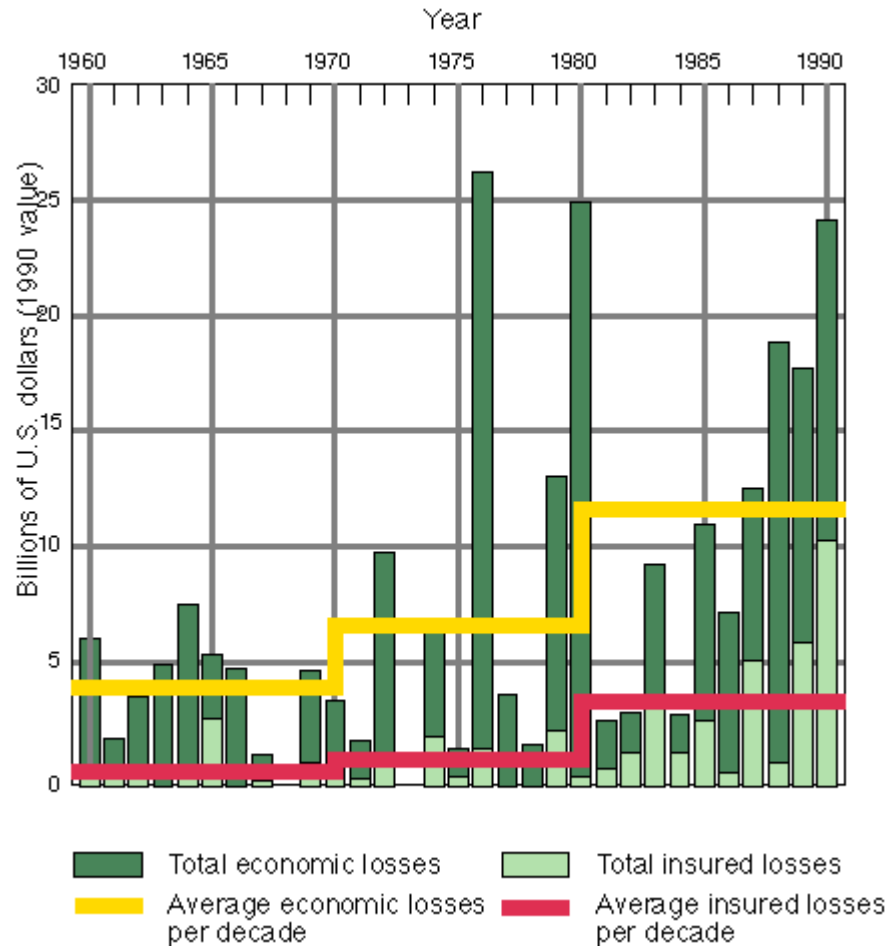
Bam Earthquake, Iran, Dec. 2003



- 60% of the deaths in the earthquakes in the past 60 years stem from the collapse of unreinforced masonry structures.
- Kashmir earthquake, 2005, 74,000 persons died, 70,000 injured, more than 3 million homeless.
- New Orleans, California, Bam, Kashmir, and many areas devastated by natural disasters use Construction systems based on empirical approaches

Natural disasters are a "growth" industry

earthquakes, floods, storms, heatwaves and droughts

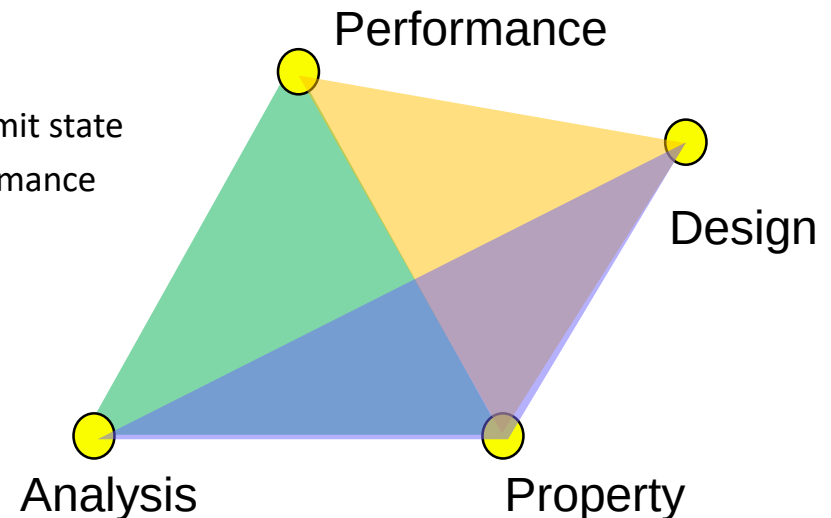


(a) Since the 1960s, economic losses from natural disasters on a global scale have tripled, while insured losses have quintupled. (after Berz, 1992, *Natural Hazards*, 5, 95-102)

(b) Data on natural disasters from the Emergency Disasters Database (EM-DAT) database of the Centre for Research on the Epidemiology of Disasters (CRED).

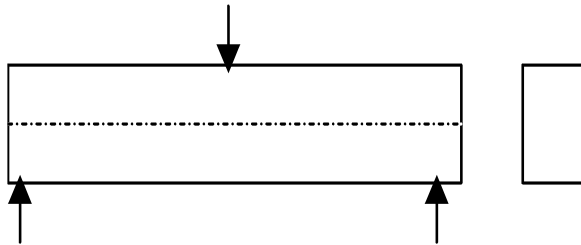
Structural Systems Engineering

- Difference Among Architectural Work, Structural Engineering, and Construction
 - Architects work with the conceptual design and planning
 - Structural Engineers address the components of the load bearing, safety and functionality of the building. Member sizing, columns, beams, frame, connections, and methods of carrying the loads applied to the structure in a safe manner.
- Structural Engineering Tasks
 - Concept Development
 - Analysis tools, math, science, materials chemistry, physics- mechanics-dynamics, building physics
 - Design development- Structural Analysis, Structural Design, Construction know-how
 - Construction Methodology
 - Innovative Materials
 - Structural Failures, serviceability and Experiences, limit state
 - Natural and Human environments service life performance

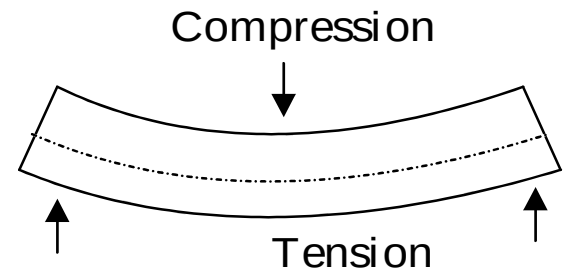


Analysis & Design

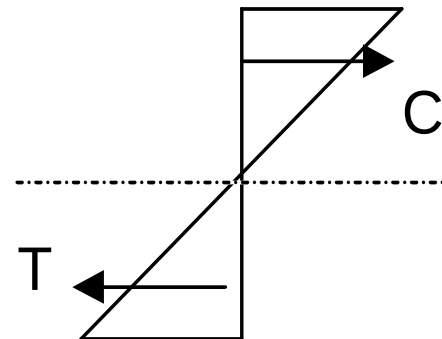
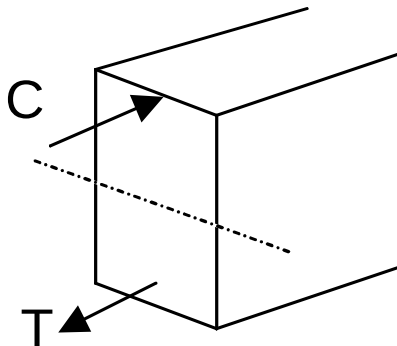
- A load applied to a beam will cause a deflection. The deflection is the means of the beam to act as a spring to store and retain the energy imparted to it. This available energy is accessible for causing damage if its magnitude exceeds the strength of the beam.



Beam Before Loading



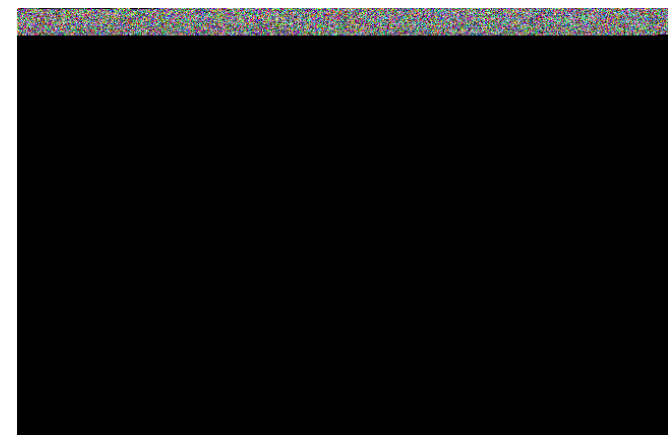
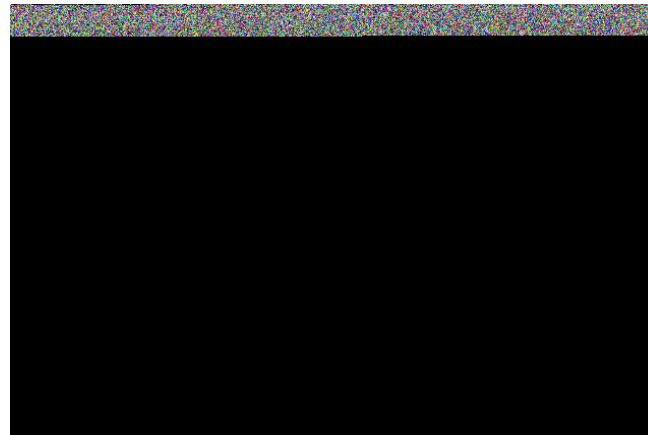
After Loading



Stress Analysis

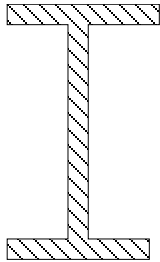
Structural Engineering related Subjects and specialties

- Undergraduate Program
 - Statics
 - Dynamics
 - Deformable Solids
 - Structural Analysis
 - Steel Structures
 - Reinforced Concrete Structures
 - Wood & Masonry Design
 - Matrix Methods of Structural Analysis
- Graduate Program
 - Structures- analysis
 - finite element method, stress analysis
 - Structures-design
 - advanced steel & concrete design
 - Structures-mechanics
 - Structures-materials
 - Structures-geotechnical
 - Structures-seismic

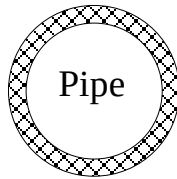


Structural Shapes

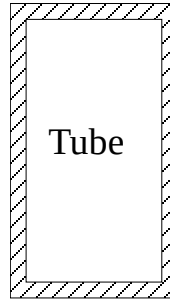
Wide Flange Beam



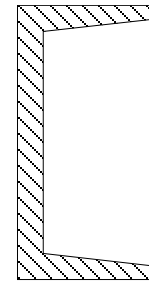
Pipe



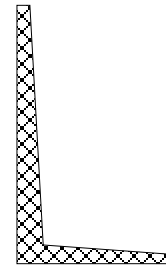
Tube



Channel



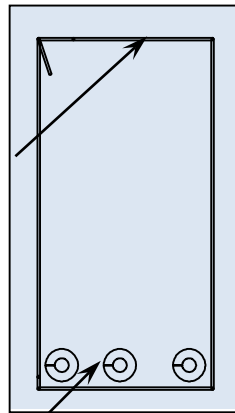
Angle



Stirrups

Main Bars

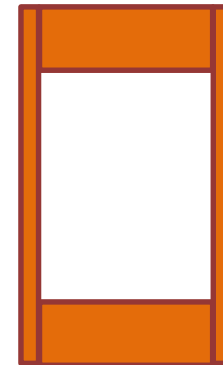
Reinforced
Concrete



Glue Laminated
Wood Beam



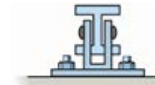
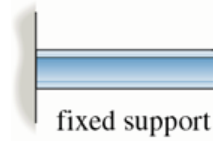
Plywood Box Beam



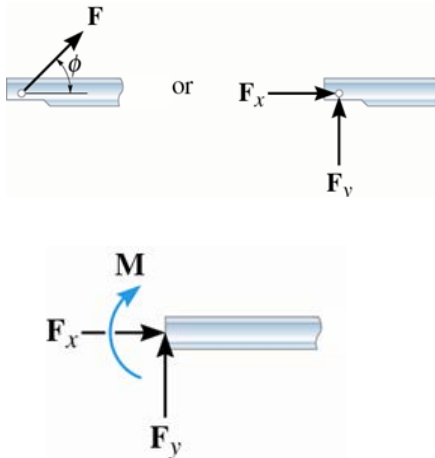
CAD, Structural Analysis and Design (FEM), BIM



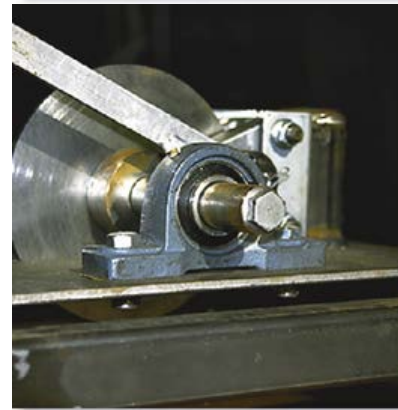
Support Conditions



pin



Ball and socket joint



Bearing support



Pin support



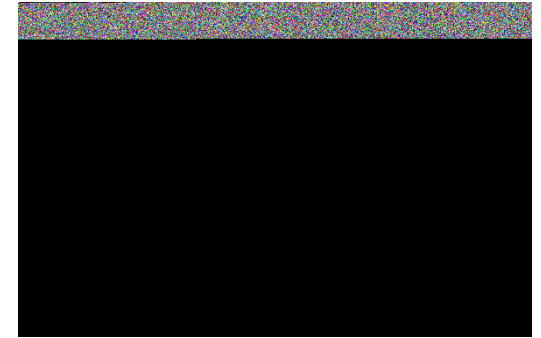
Flange Welded



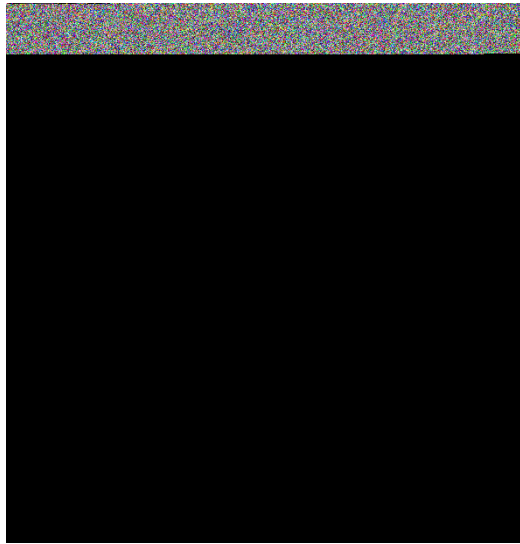
Flange Plate Bolted

Wood, Cold formed steel applications, Concrete, hybrid systems

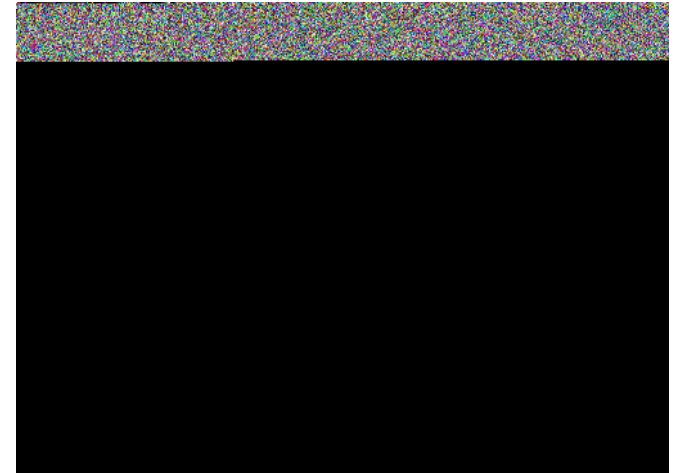
- Engineered floor joists
- Floor Trusses
- Framing
- Prefabricated elements
- Interior and exterior panels



Hurricanes & Earthquake Events



Andrew, 1992, \$25B estimated damage



Katrina, 2005, Levy Breaches –flood
Structural damage, \$200 billion



• Northridge, CA, 1994
Estimated damage, Over \$10B

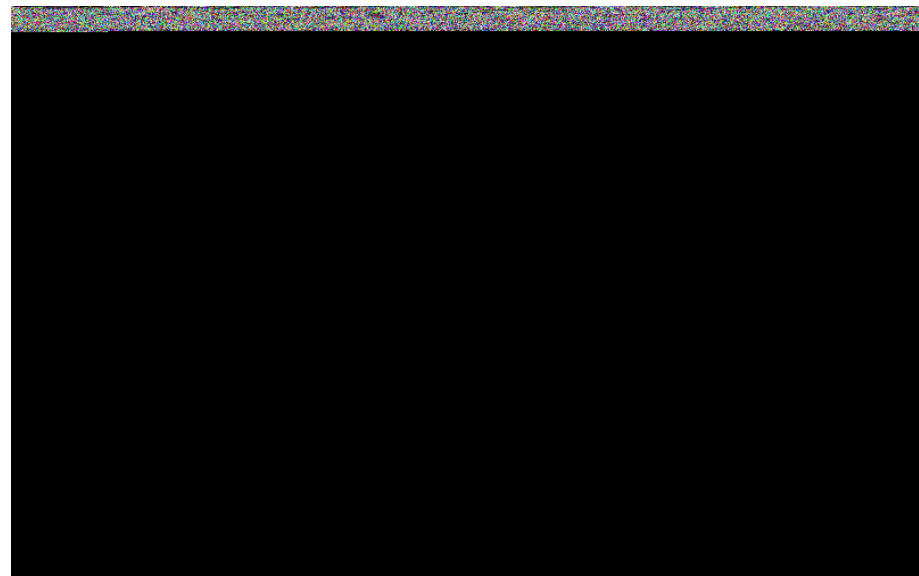


Kobe, Japan, damage at \$310 billion



Taiwan, 2011

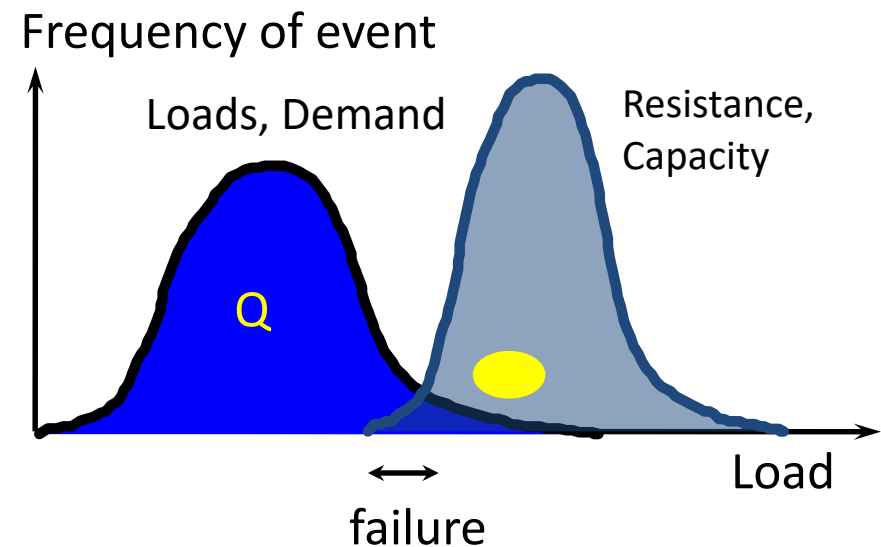
Building Codes- Municipal, ASCE, ACI, AISC, Masonry, Wood



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Load and Resistance Factor Design

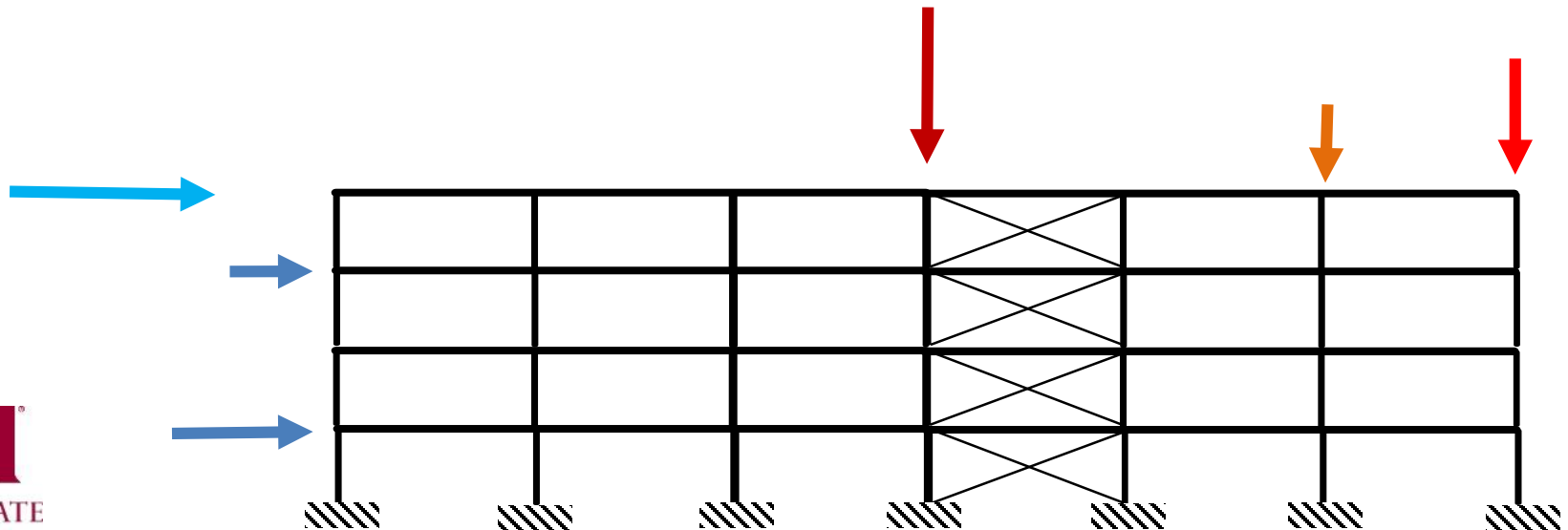
- Theoretical strength, R_n
- resistance over-capacity factor, ϕ_n
- Structural resistance = $\phi_n R_n$
- Resistance > All applied loads
- $\phi_n R_n > \lambda_i Q_i$



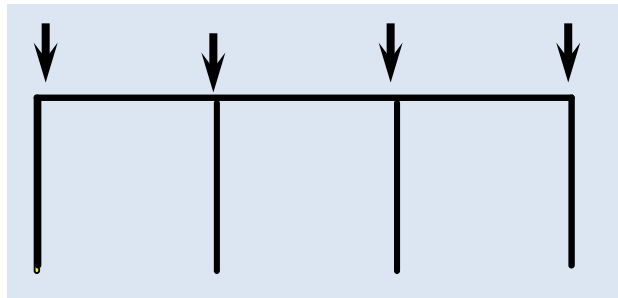
Design of High Rise Building

- Lateral load resistance path
- Stability and sway bracing
- Load path redundancy

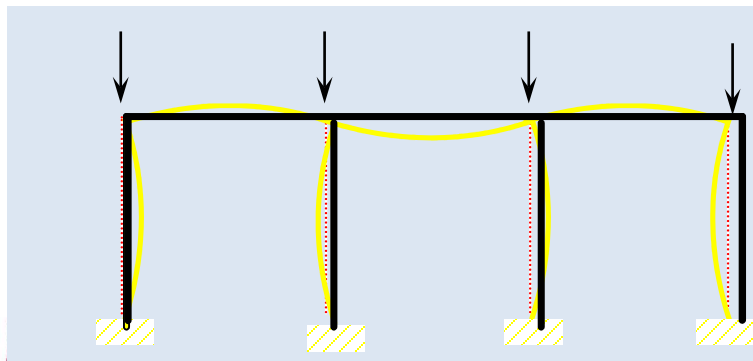
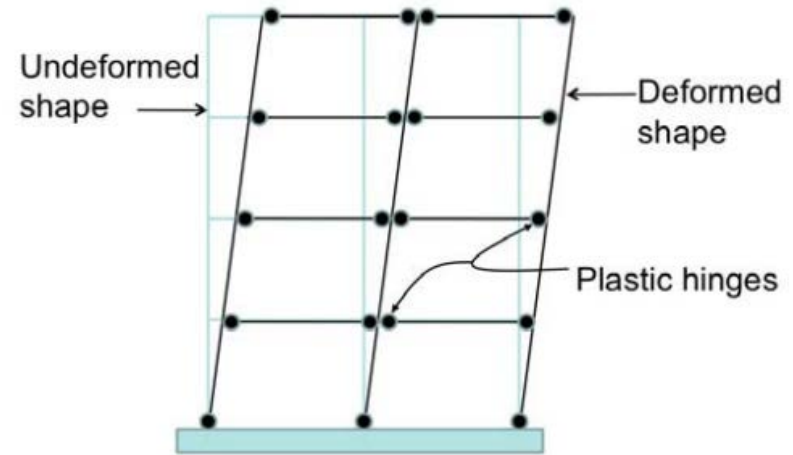
	structural design		
Loads	Primary	Secondary	Tertiary
Dead load			
Live load			
Wind load			
Snow load			
Seismic load			
Fire load			
Impact load			
black Swan			



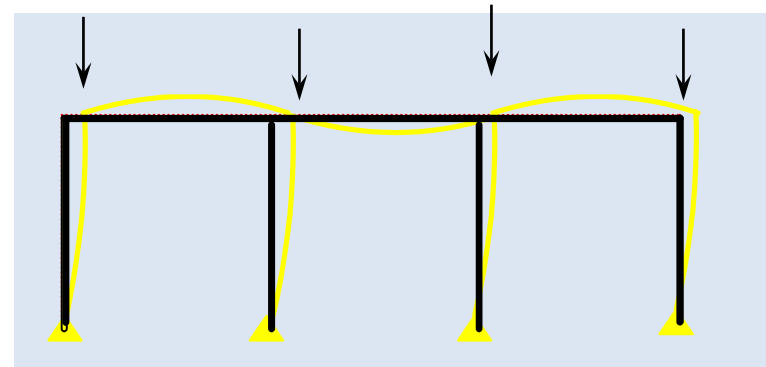
Failure, Strength, Stability, Redundancy of design



Original structure



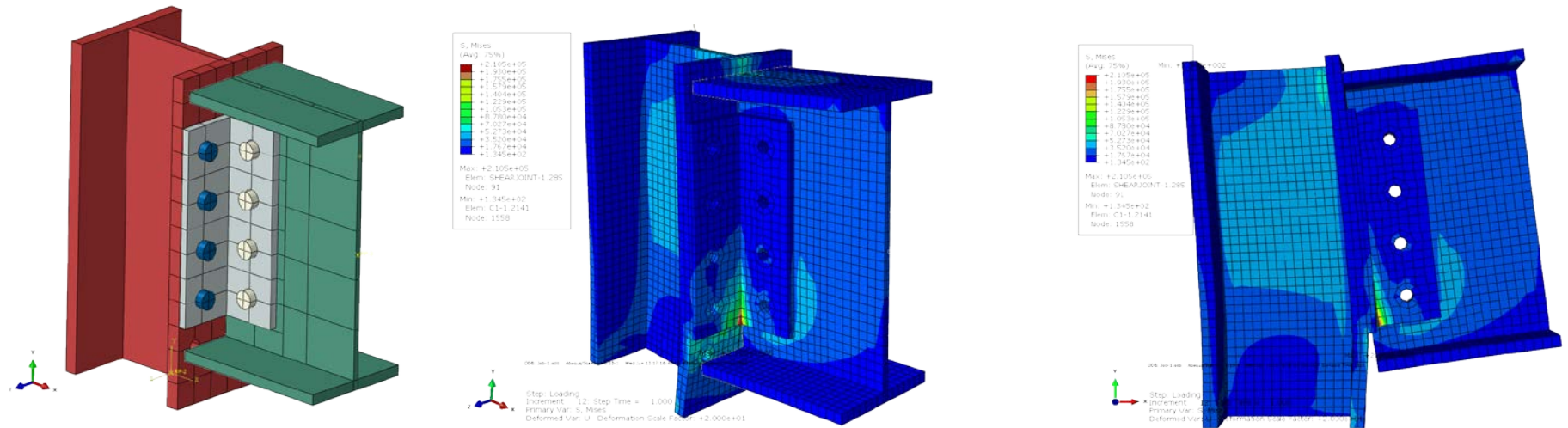
Side sway prevented



Side sway allowed

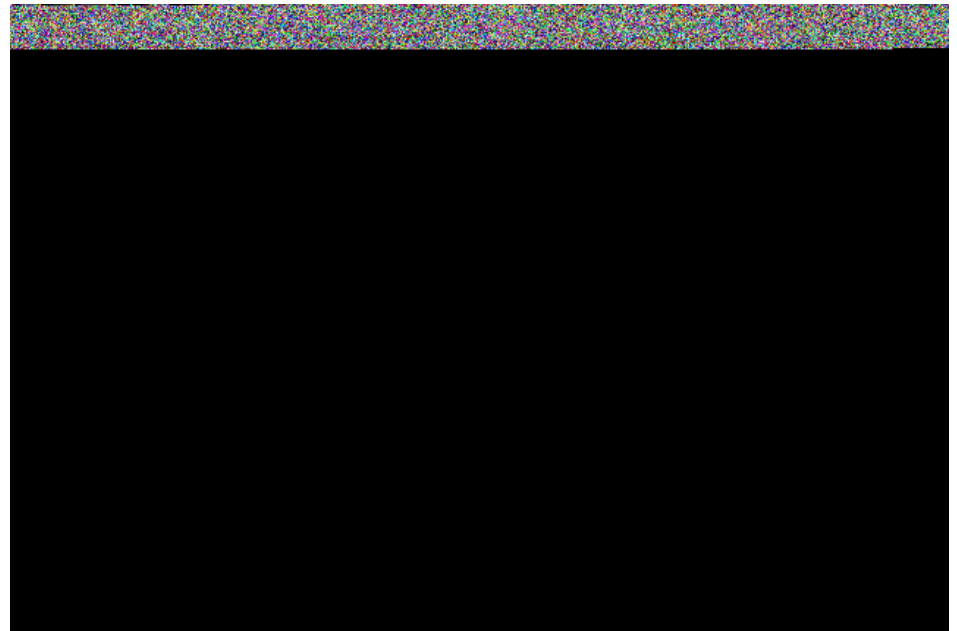
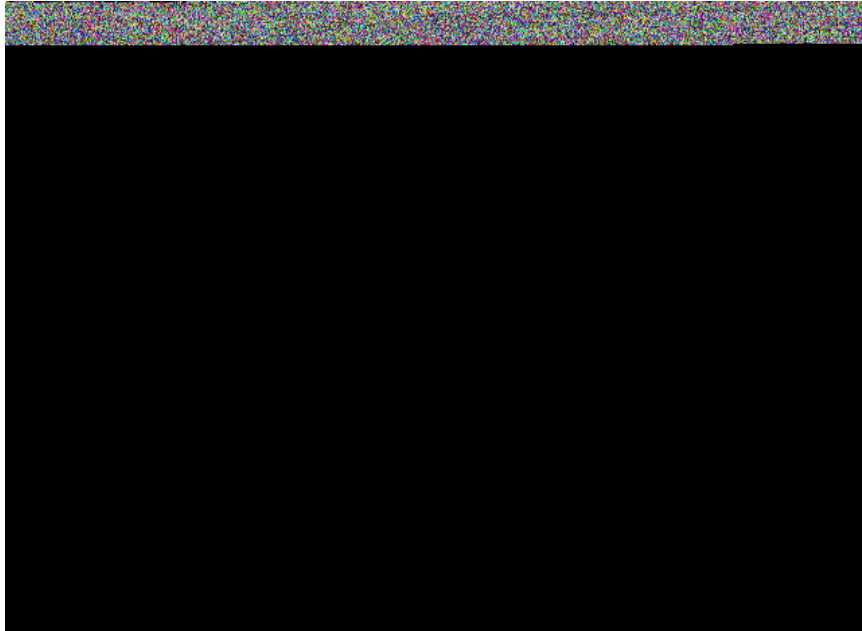


Finite Element Modeling

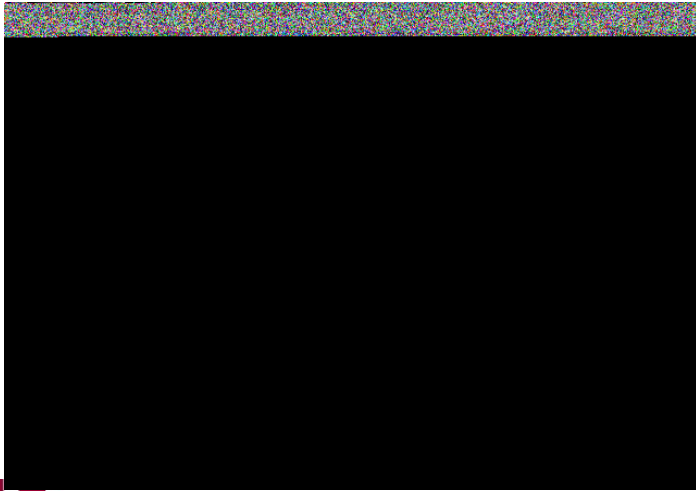
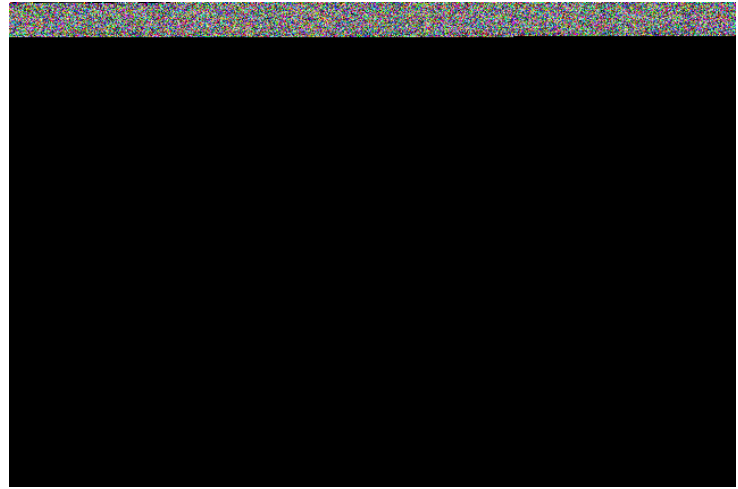
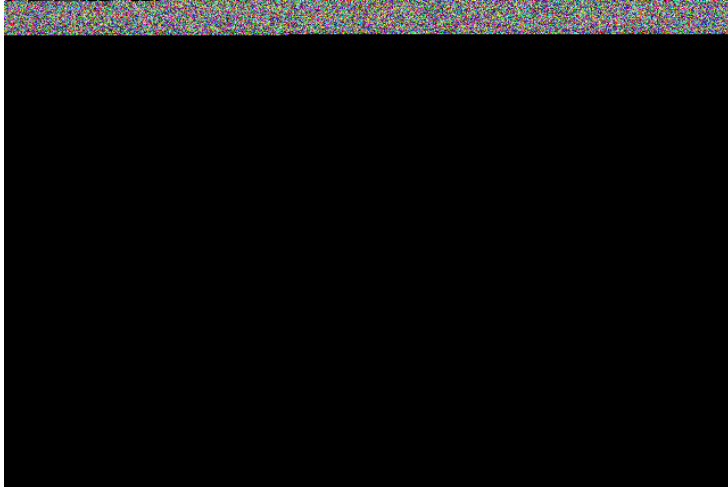


Materials, Mechanics, Structural Analysis, Structural Design, Serviceability
Design based on Structural reliability, robustness,

Cantilever Connection, lacks redundancy



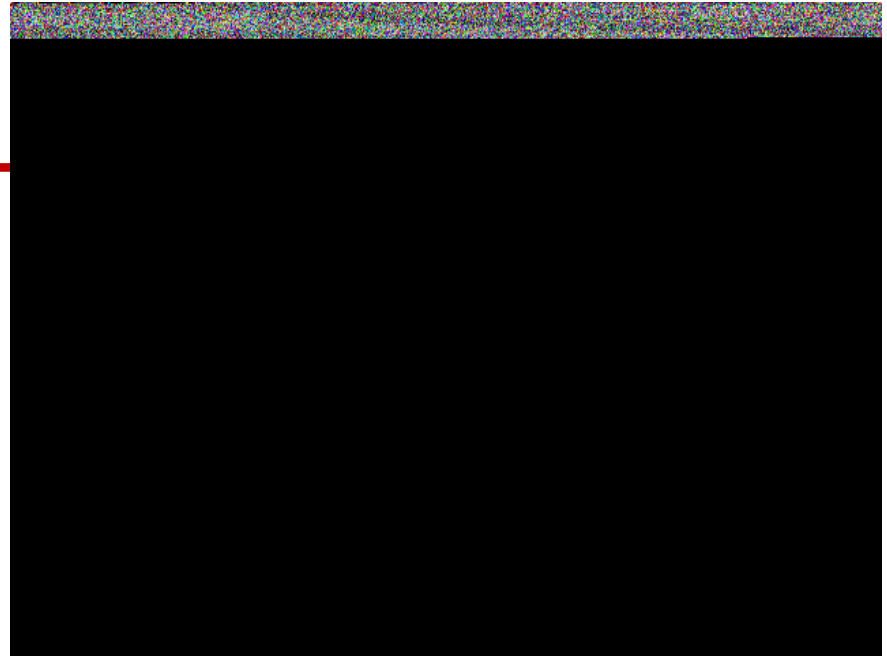
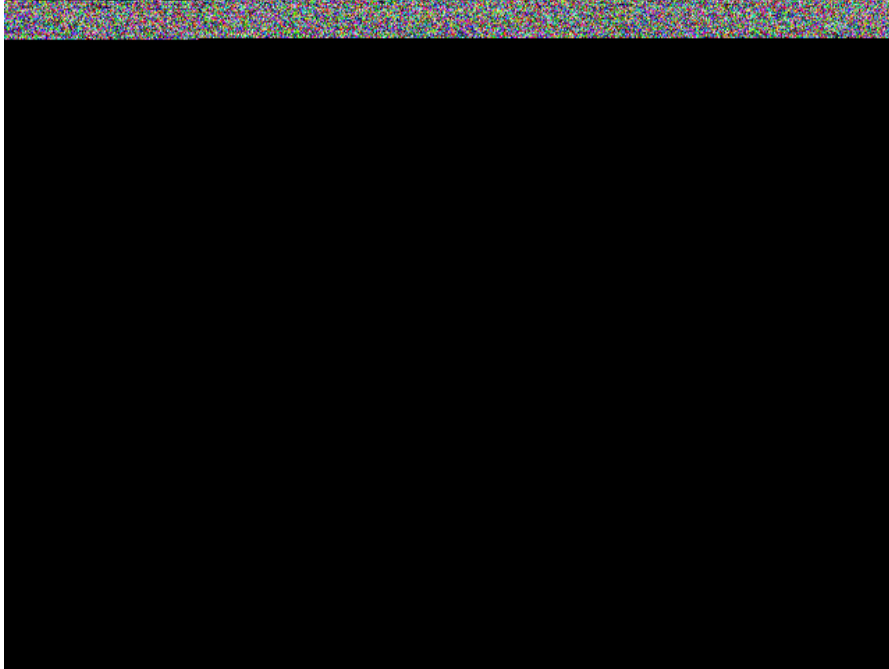
Truss systems-lack redundancy



Drill, Weld, Cut



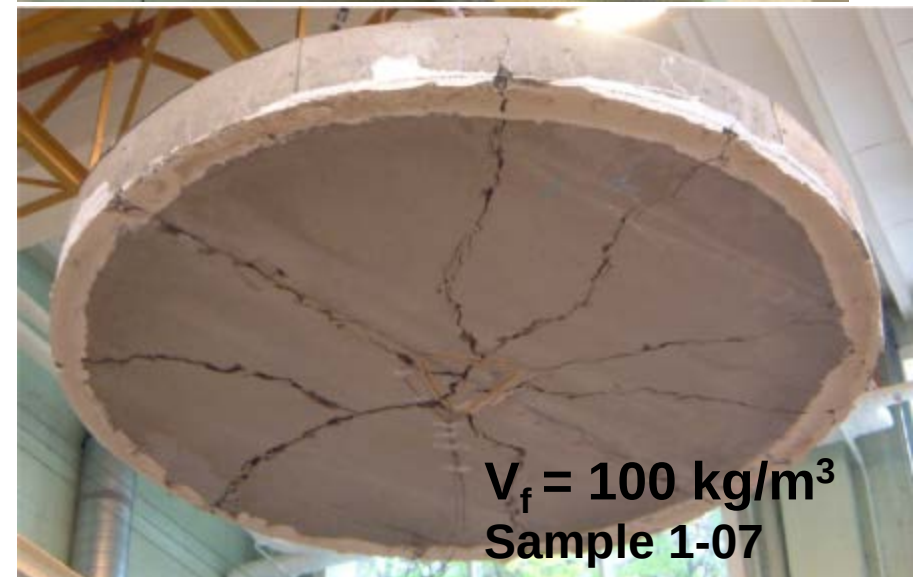
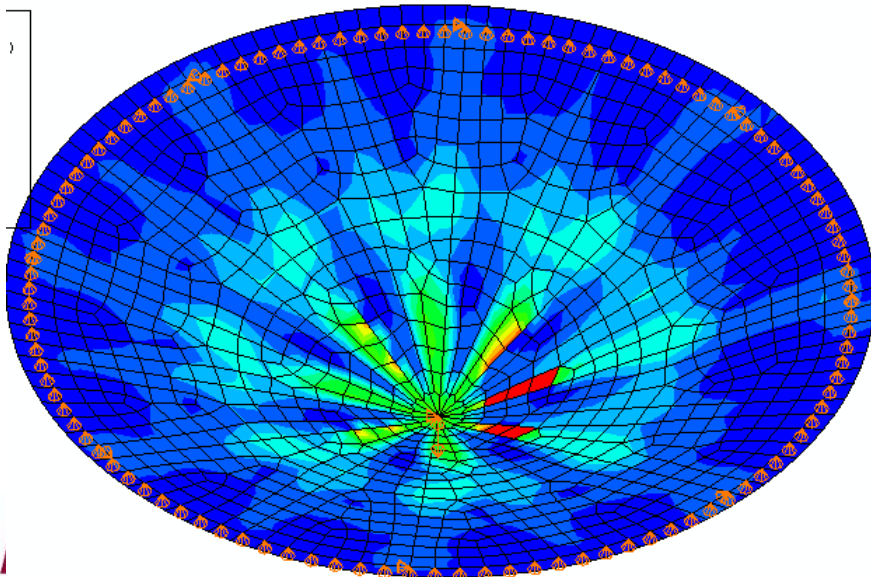
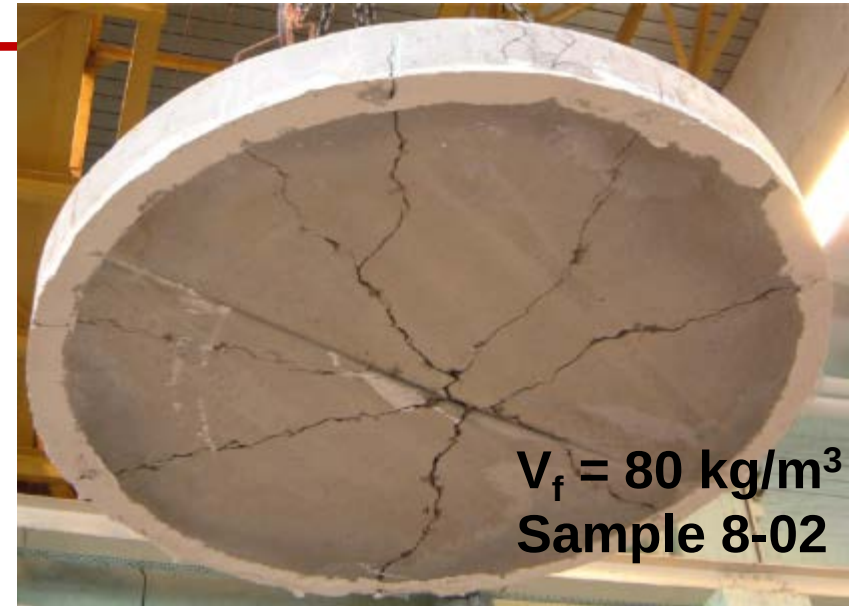
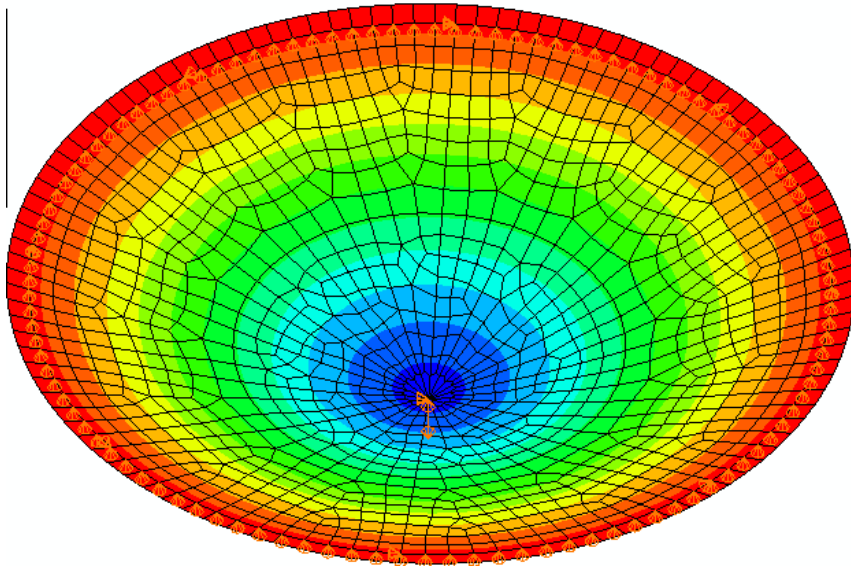
Pin, or Roller Connections



Innovations, new Design approaches, Serviceability

- Structural modeling
- Structural design with an understanding of the risks involved
- Durability vs. Strength
- Ductility, Energy absorption
- Service life, fatigue endurance
- Life Cycle modeling

Finite Element Simulation of load deformation response and Crack Pattern formation



Testing determinate and indeterminate panels



Leak before break mentality- Use the redundancy of the structure to cause multiple paths for transfer of the load



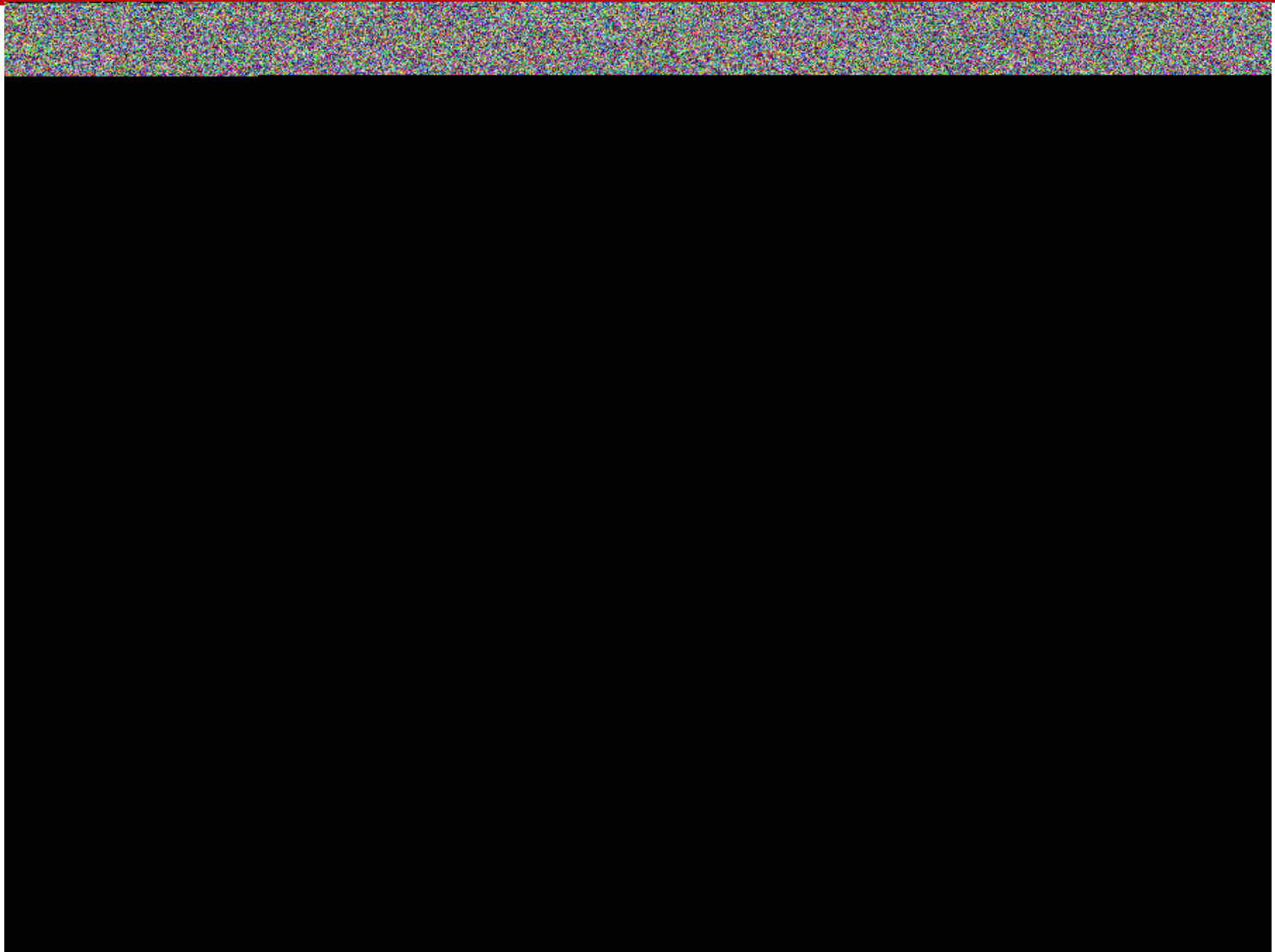
Precast Components and systems



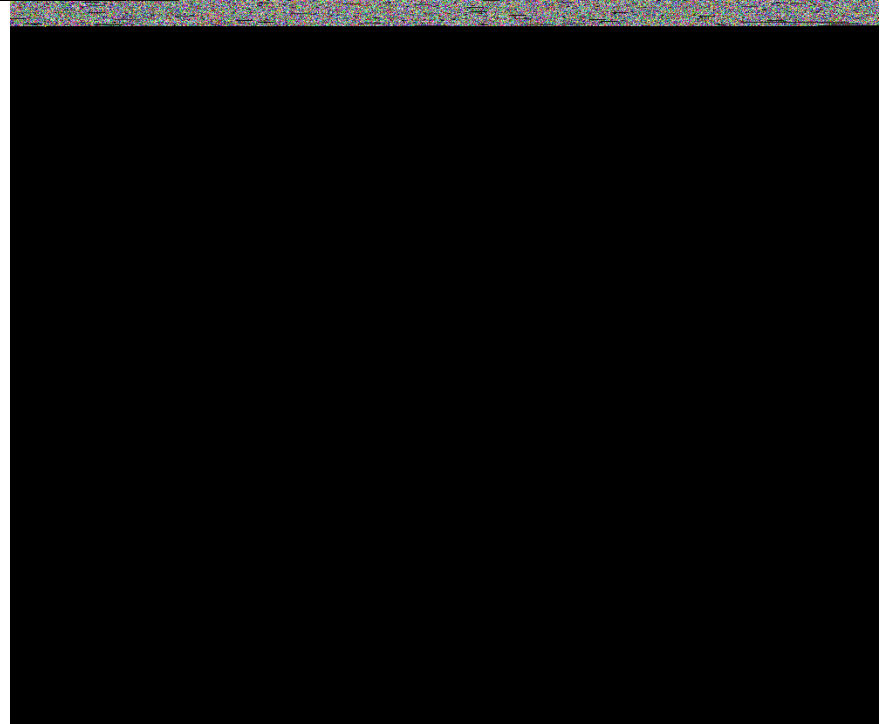
New Light gage Steel sections,

Cold-Formed Metal Framing

ASU ISTB7 Building GFRC, Exterior cladding (3/4" thick cement composite about 6" off the exterior of the building)



Design based on Redundancy, alternate load paths



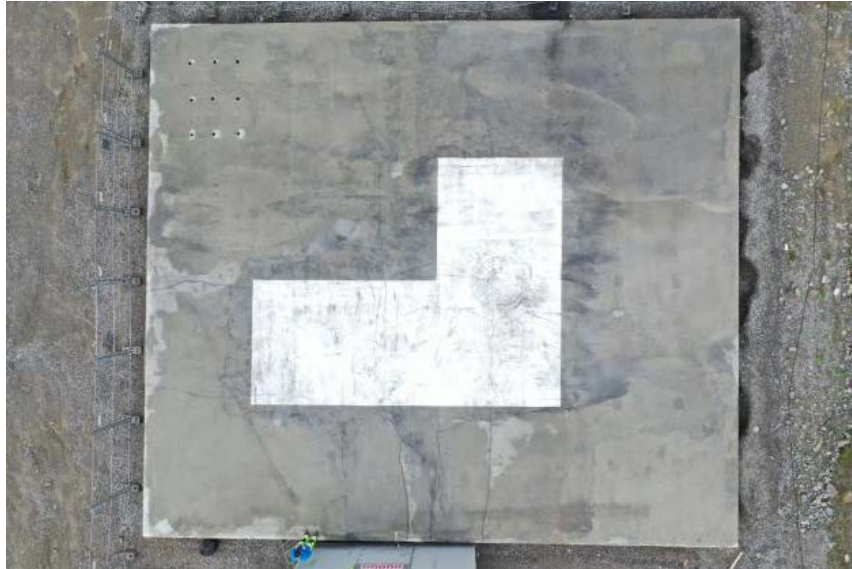
Structural Engineering Design Validation via full scale testing

- Warehouse floor, 8" thick concrete slab with steel fibers, Full scale tests to assess the capacity- 4000 kg/m^2 equal to 820 psf



Parking structure design requirements, 40 psf of live load. The top level needs Live plus snow load 54-60 psf of total live load for the roof level

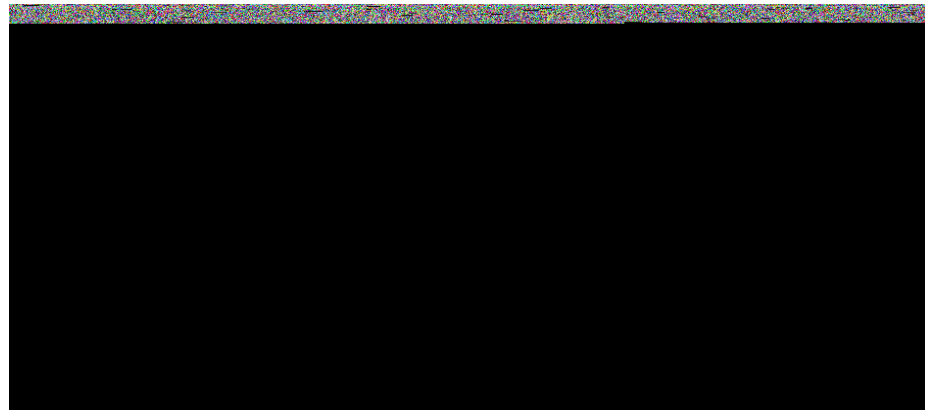
Homogeneous slab on piles



Capacity in excess of twice the design load due to redundancy of load paths

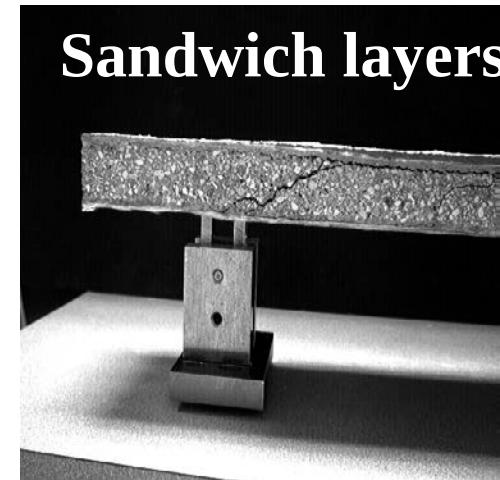
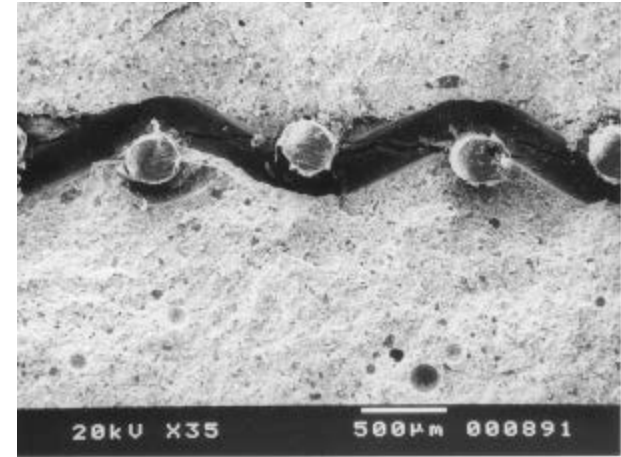


WTC 5 and WTC 7

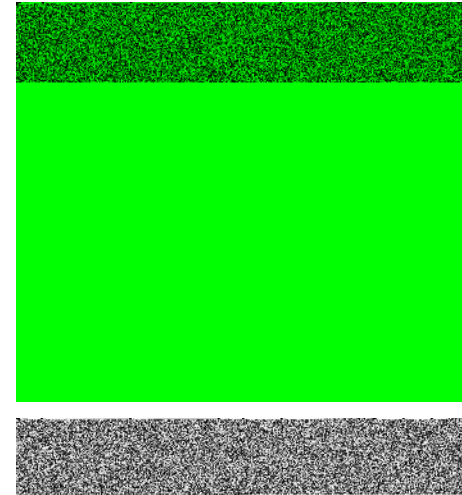
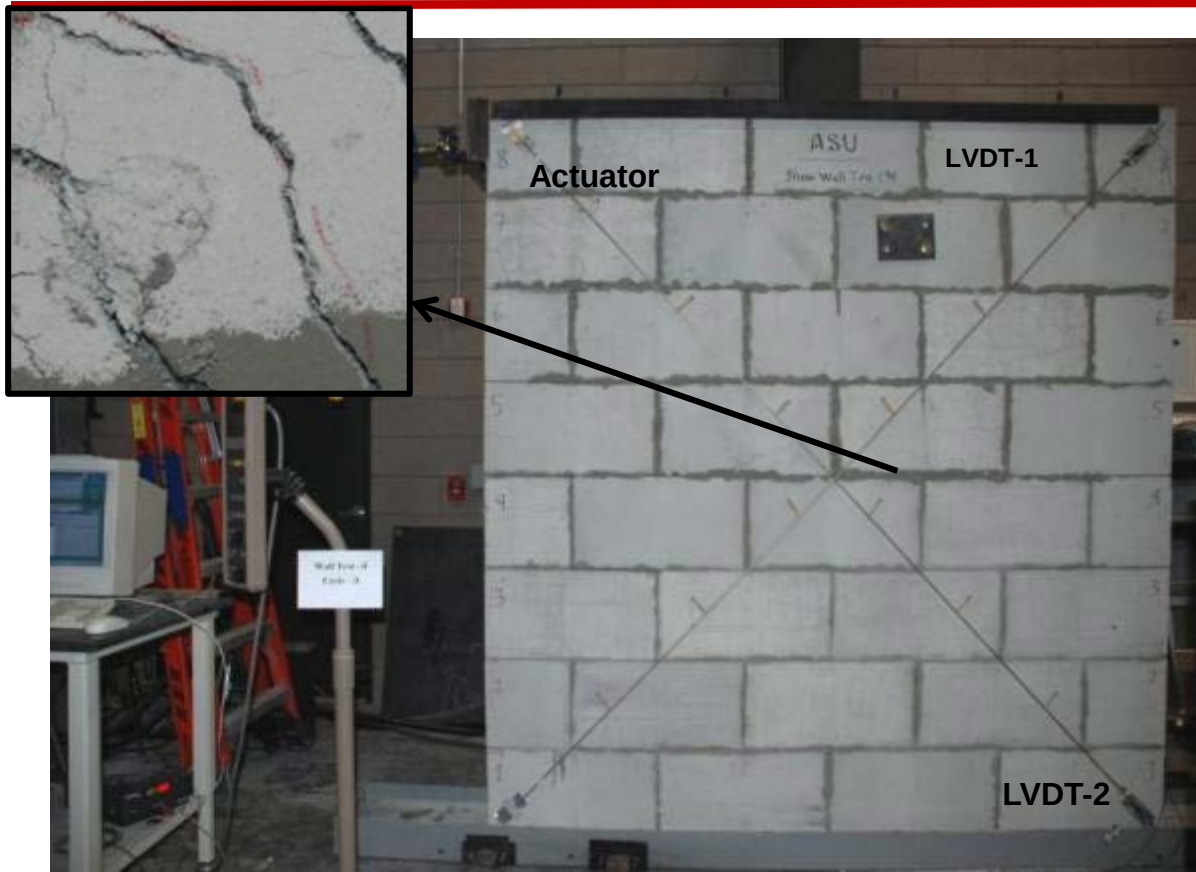


High Performance Fiber and Fabric Reinforced Concrete- Thin Sections and shapes

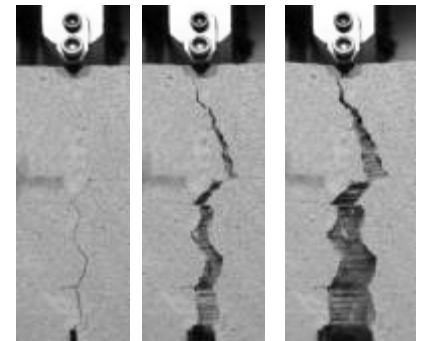
Low cost equipment set up
Uniform production
high performance fabric-cement composites
Tension, Compression, beam members
High pressure pipes



Thermally efficient wall members using aerated concrete



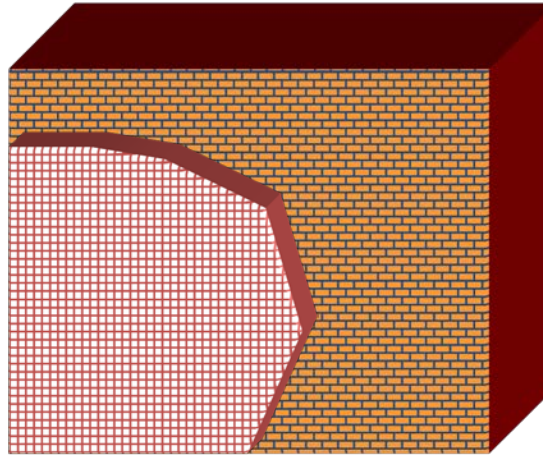
- Durable, Energy efficient, Effective R-value, 3.4/in,
- Easy to use, Fire, Pest resistant, no autoclave
- Acoustical insulation, impact resistant



SCC and Fiber composites Construction-Cast in place SFRC



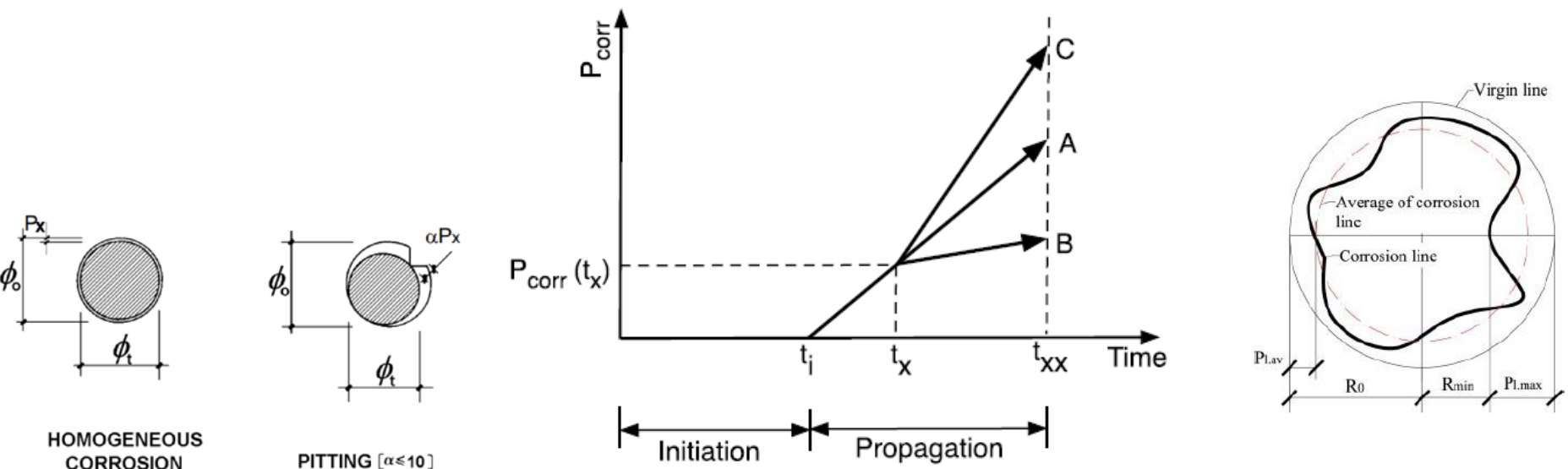
Repair of Unreinforced Masonry Walls Large scale tests



In-plane shear tests
to simulate seismic action.

Standards for Durability based Design and Repair

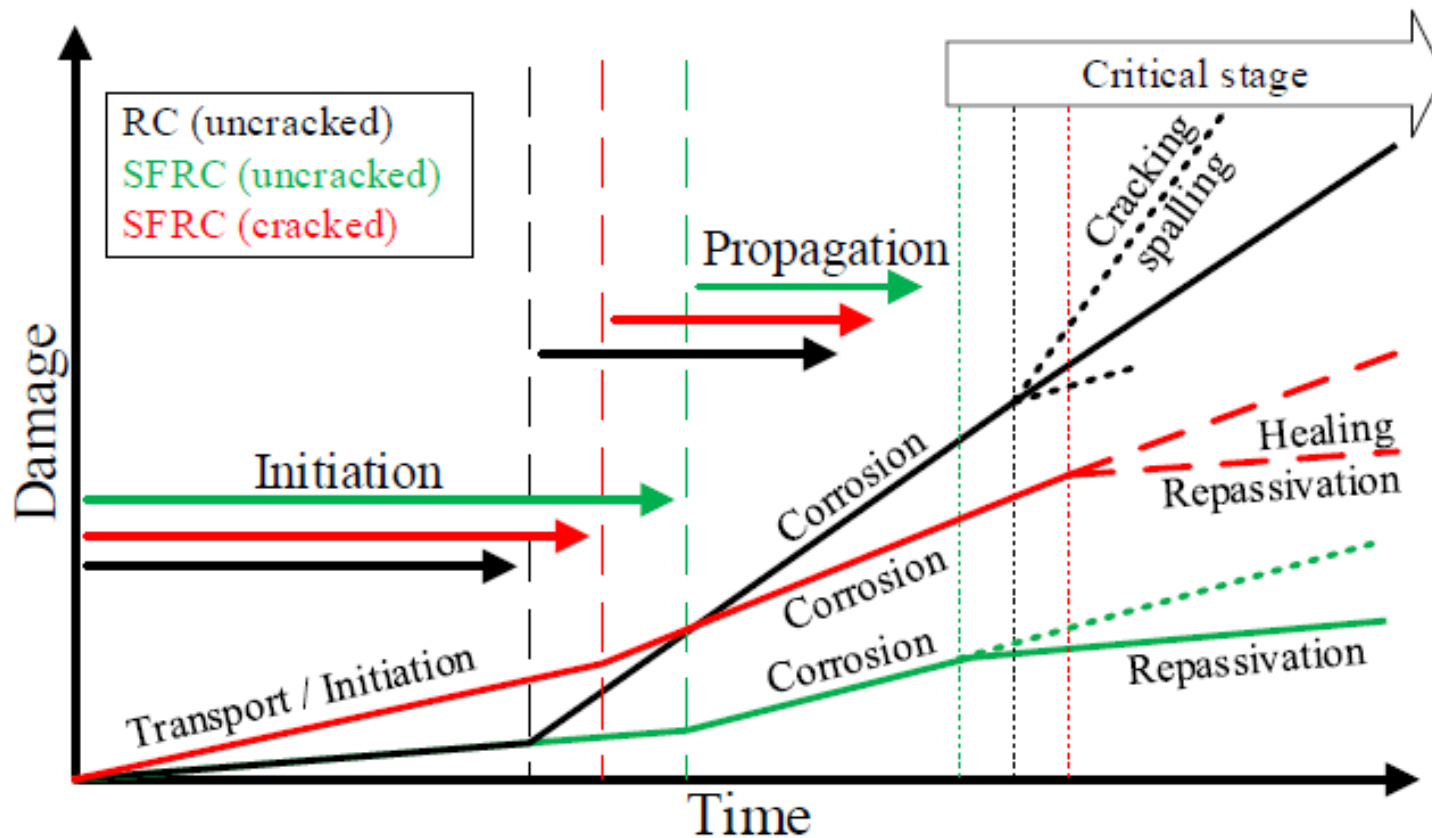
- **Carbonation, Chlorides, Sulfates, Chemical / acids in Concrete**
- Carbonation, reaction with CO_2
- Chlorides, lower pH promote Corrosion of rebar
- Sulfates, excessive expansion and internal damage
- The initiation phase is governed by the time to depassivation of the steel due to reduction of the pH in the pore solution of the cement matrix (e.g. carbonation or acid attack) and/or by the ingress of halides (e.g. chloride ions) through the cement matrix or cracks.



Exposure Assessment methods

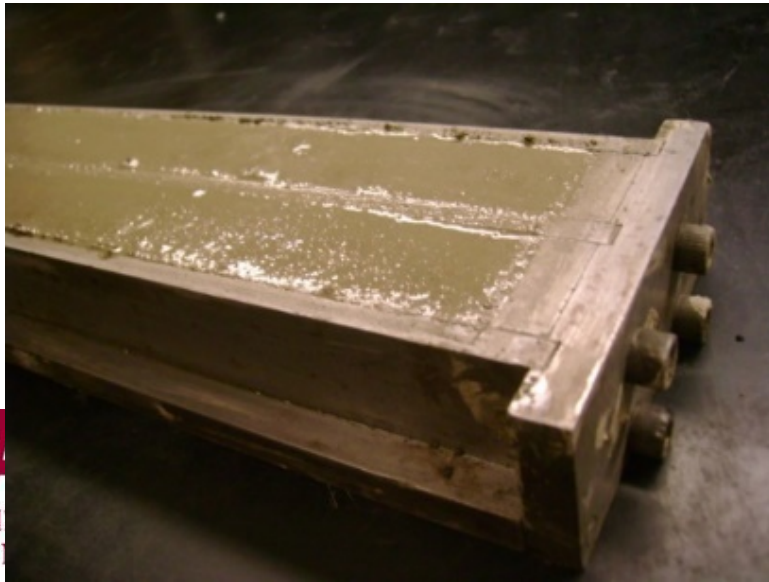
- Empirical Approaches, testing, and QC criteria are a major barrier to better design procedures.
- Exposure simulation, environment, wetting –drying cycle, exposure, electrical methods to accelerate the situation, using elevated temperature, concentrations, harsh conditions
- Mechanical tests, Visual inspection, Chemical ionic profile , wet chemistry, X-ray fluorescence
- Petrography, thin section, polished seections, SEM,
- Statistical data analysis,
- Modeling, statistical and probability based, or multiscale modeling

Corrosion Stages, maintenance, and materials effects



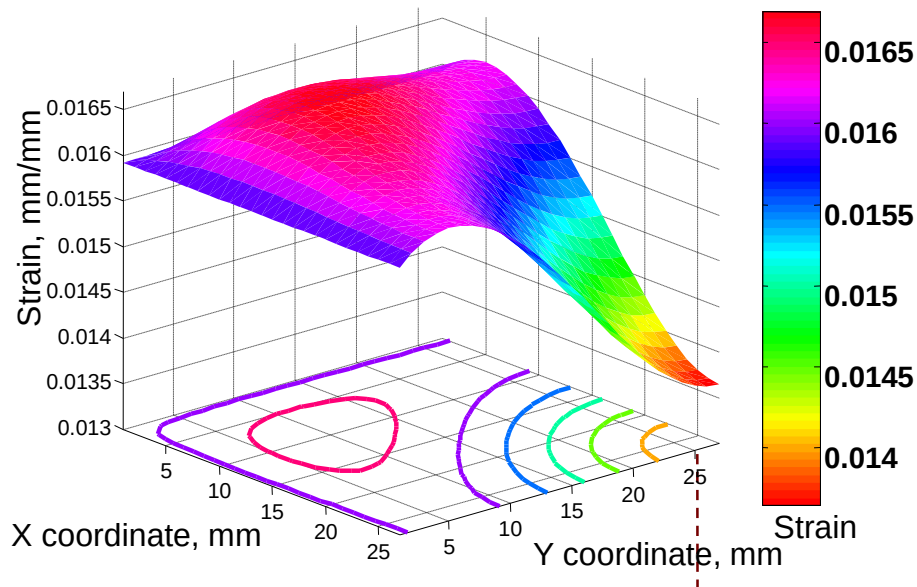
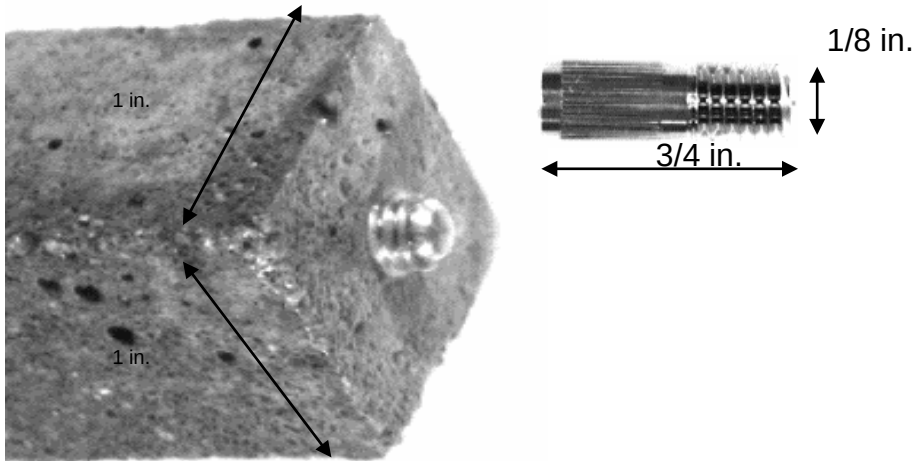
Sulfate attack

ASTM C 1012 test procedure



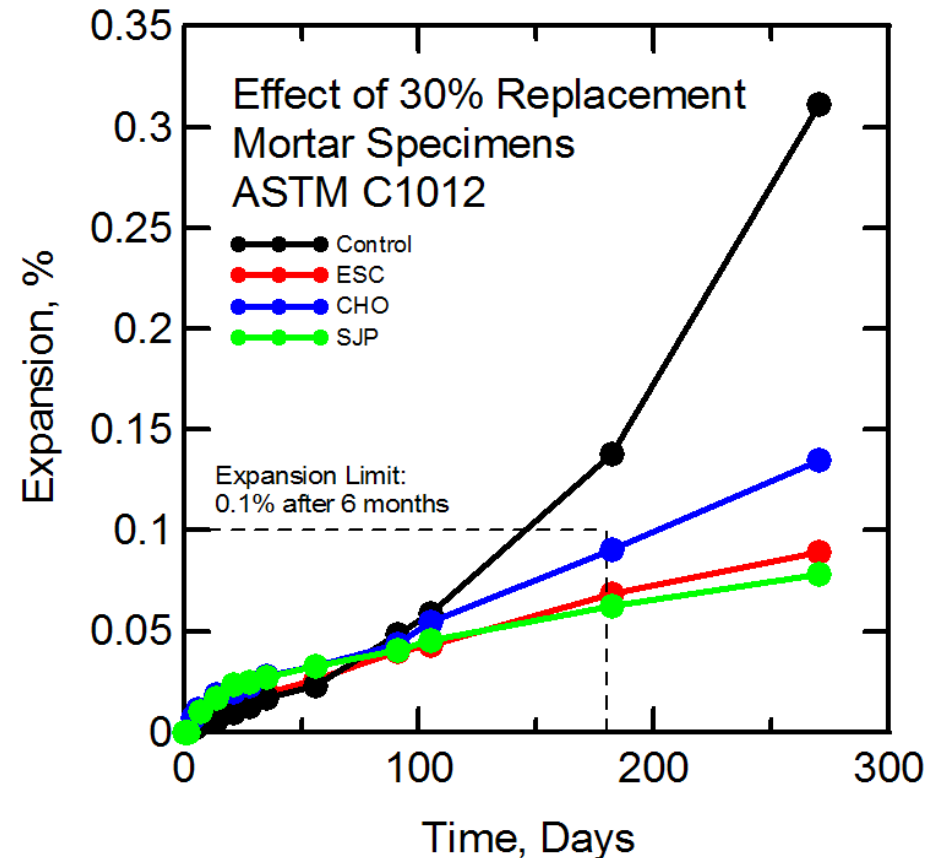
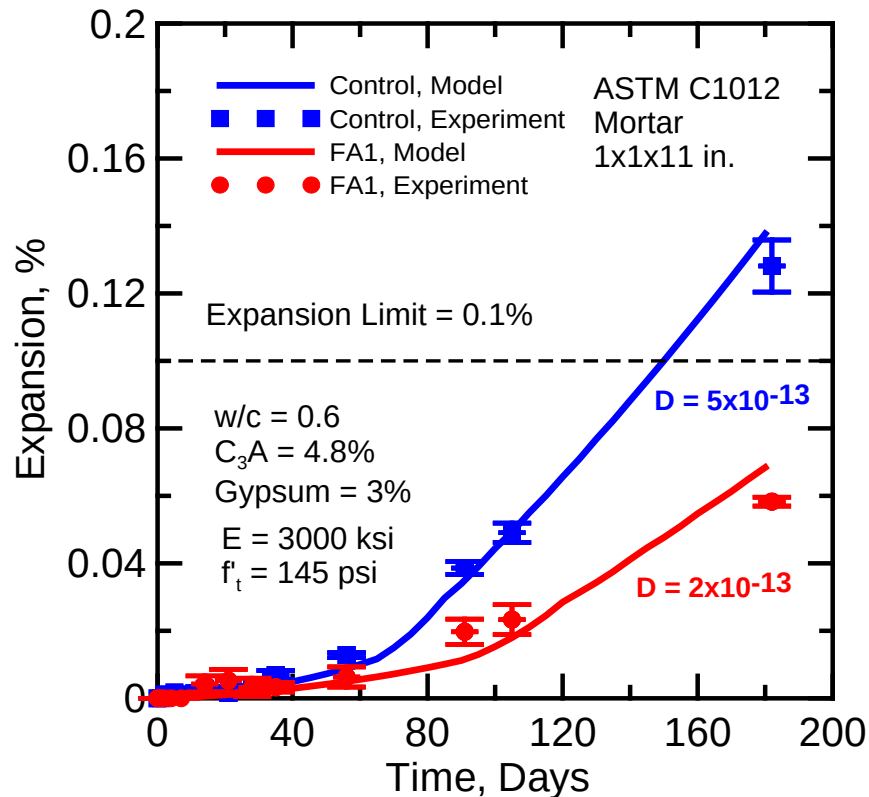
Sulfate attack

Standard test vs. modified test

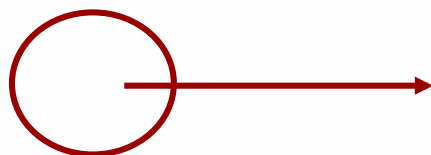
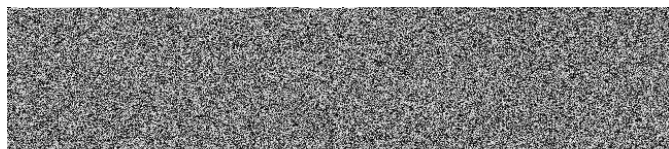


9 months expansion for standard test, Mortar specimens Control vs. flyash

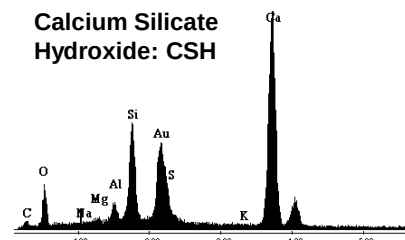
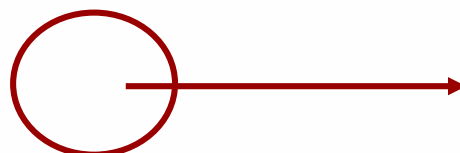
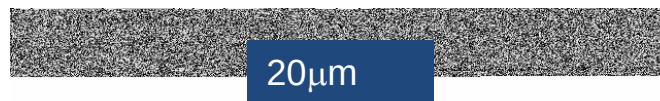
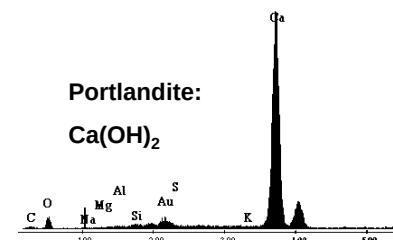
1"x1"x11"



Microstructural studies (SEM-EDS) Produced crystals

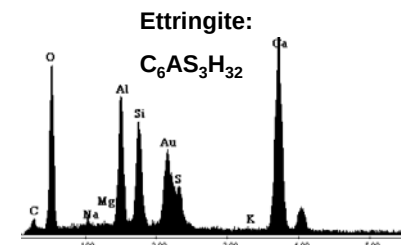


CHO-20-Center



20µm

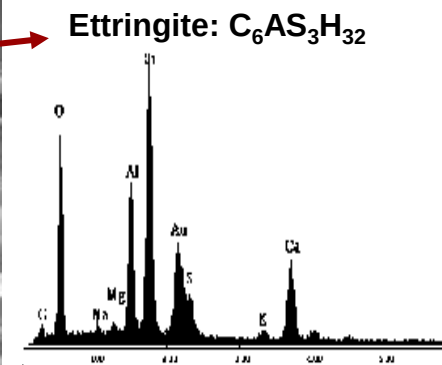
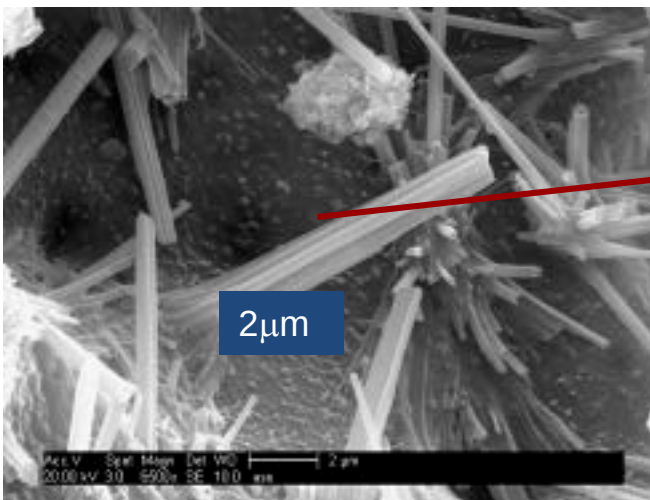
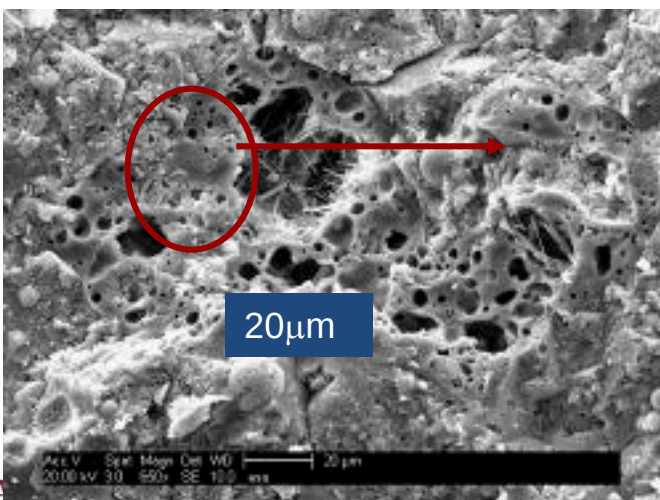
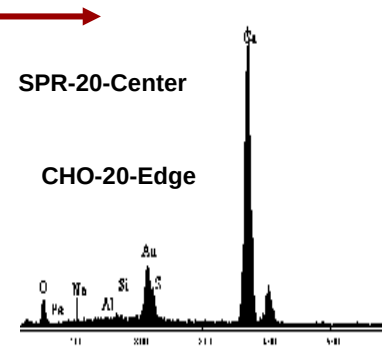
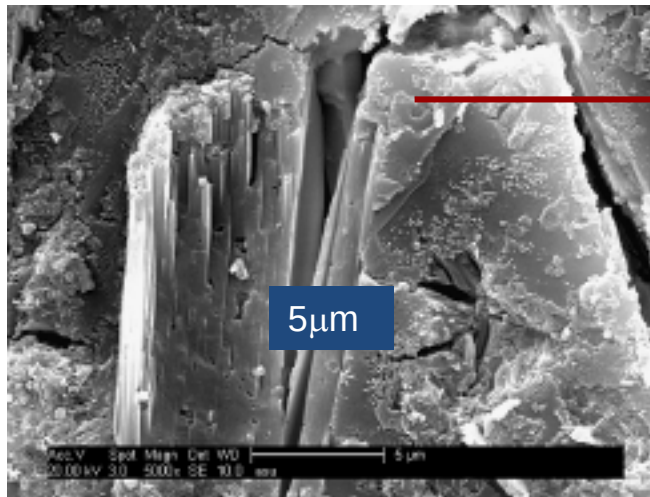
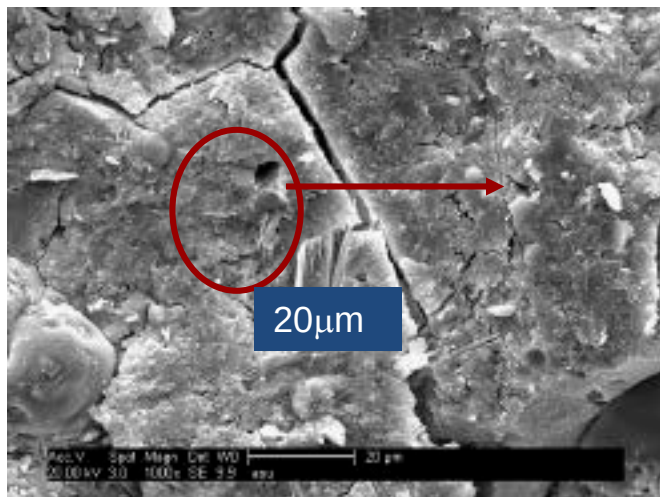
10µm



SJP-20-Center

Microstructural studies (SEM-EDS)

Produced crystals

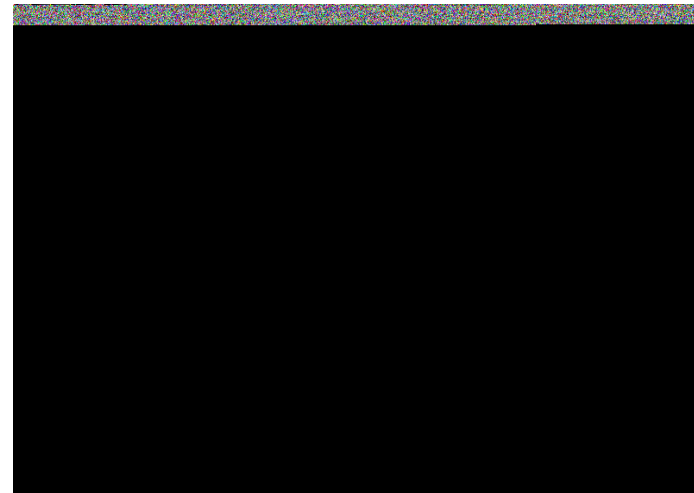


Service life modeling

- Fatigue life
 - Durability
 - Corrosion rate
 - Intermediate repair strategies
 - Freeze thaw damage
 - ASR / alkali aggregate reactions
 - Fire
- Impact and explosion
 - Strain rates for different loading velocities. High dynamic loads can be caused by:
 - impact of vehicles, trains or airplanes;
 - impact of projectiles and missiles;
 - impact of debris;
 - surface burst explosion;
 - blast caused by nearby or distant explosions;
 - explosions in rooms, including multiple reflections.

Seismic design, assessment or retrofitting

- Seismic actions impose deformations on structures. The structure then produces the internal forces and moments which correspond to these deformations on the basis of its stiffness and resistance.



Conclusions

Additional Slides

Design of weak beam-strong column systems



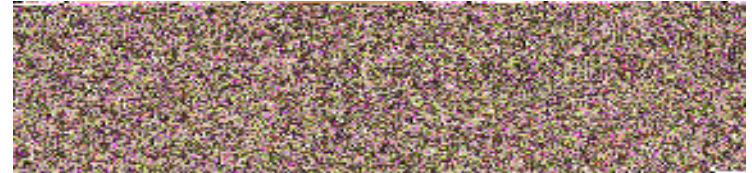
1971 San Fernando, Earthquake



Failure of Reinforced Concrete Column in San Fernando 1971 Earthquake



Failure of Reinforced Concrete Highway Structure in Kobe 1995 Earthquake



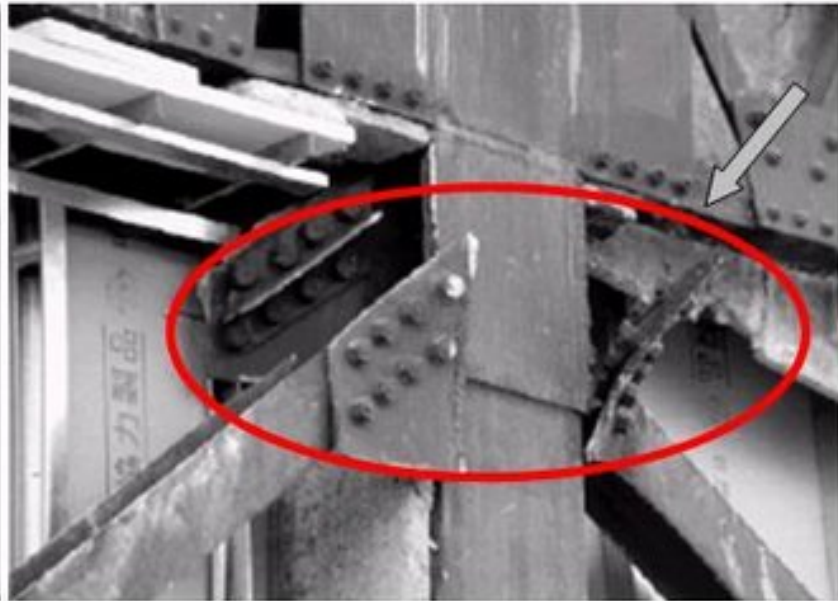
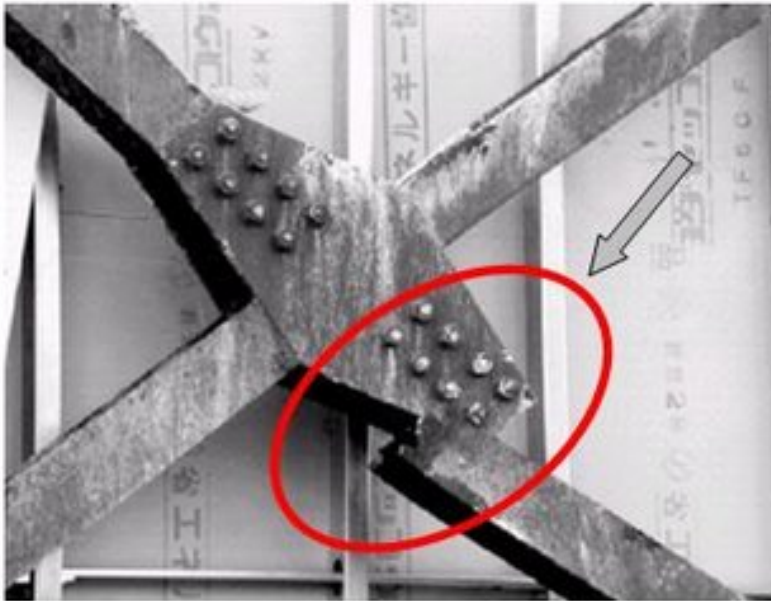
Taiwan, 1999



France, 1999, Wind



Tension and Connection Failures

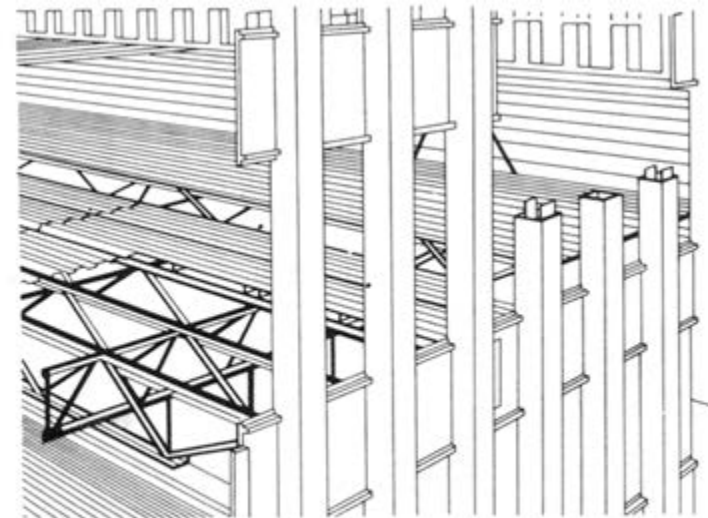
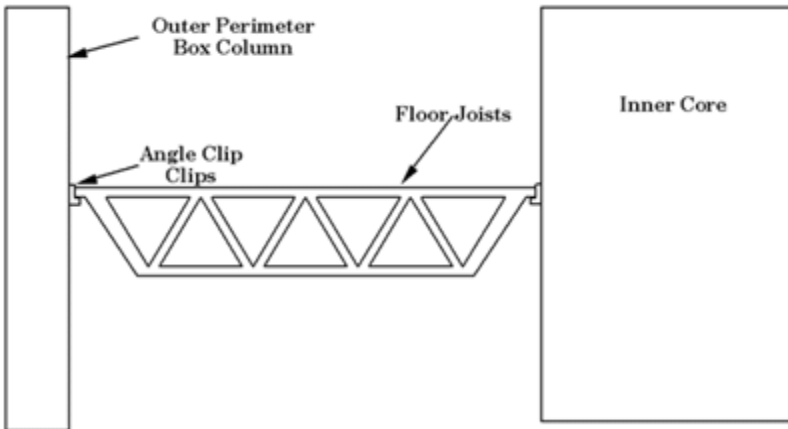


Kansas City Hyatt Regency walkway collapse

1981

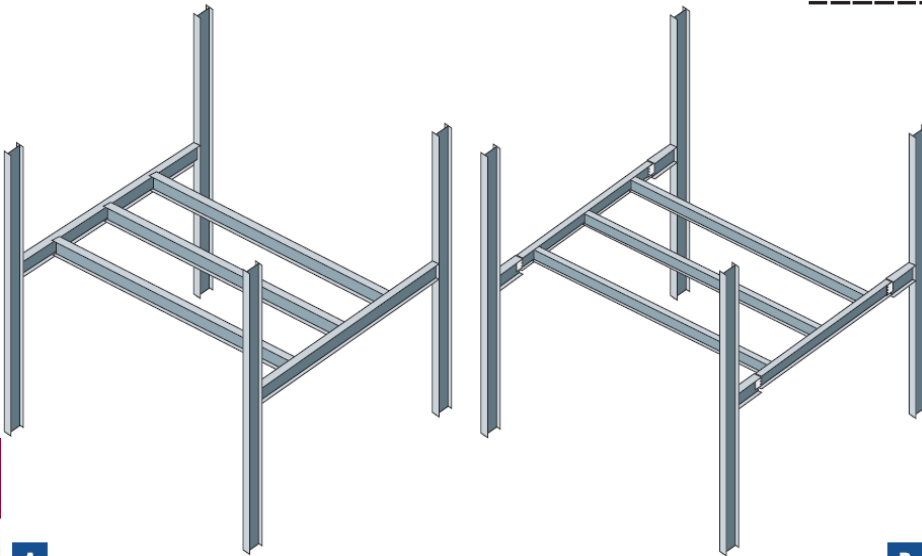
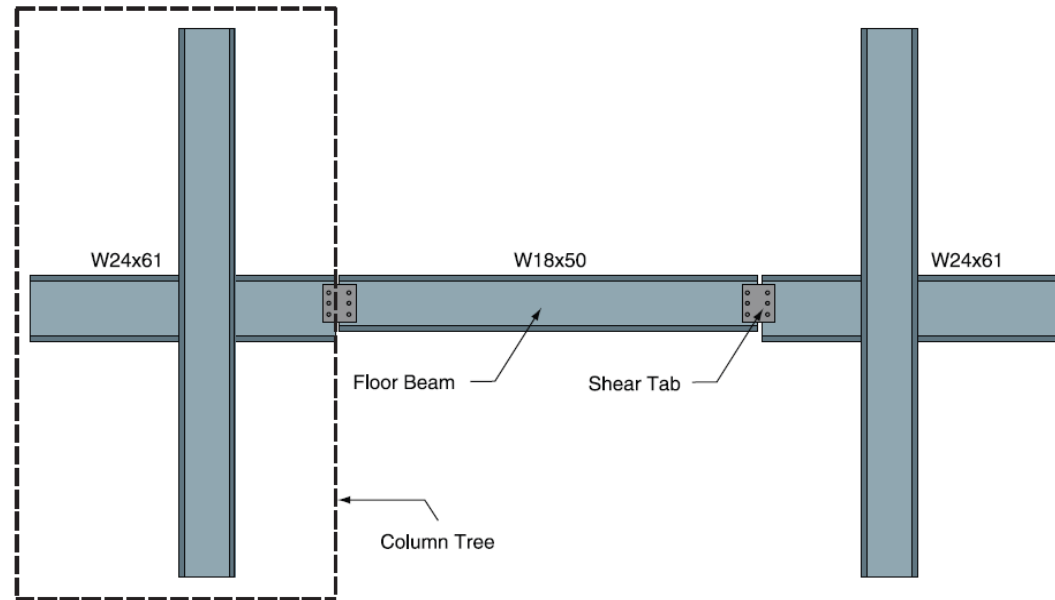


WTC Collapse



Assembly of the external wall units (alternately staggered in one-storey heights) and floor units

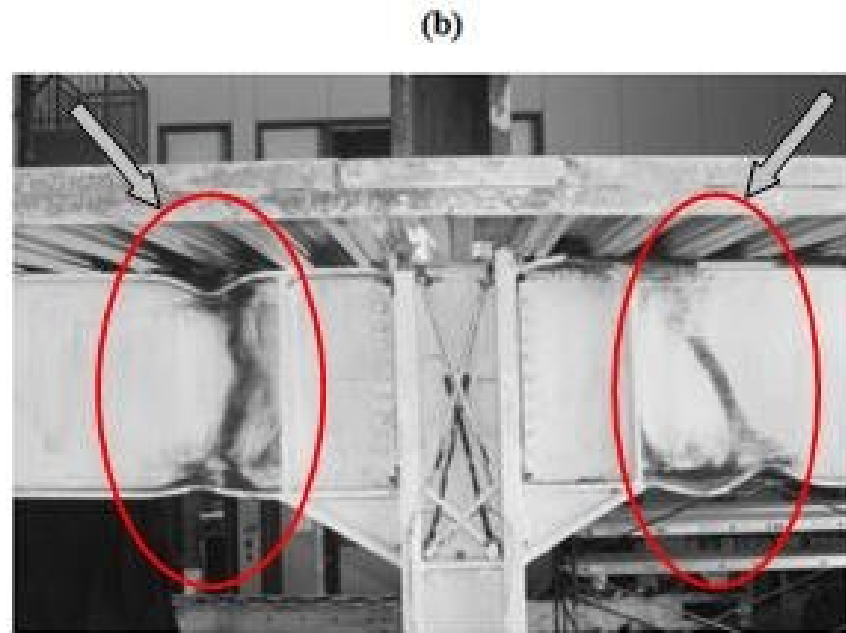
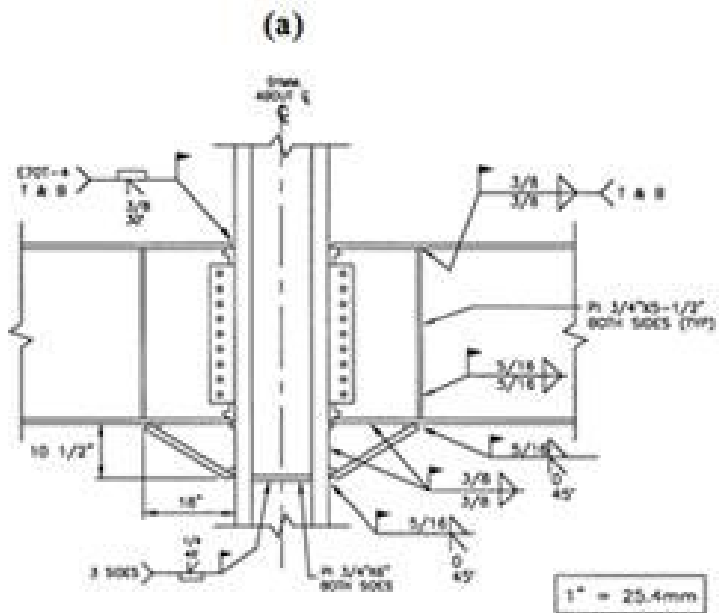
Modular Construction- Collapse of WTC-7



Plastic Hinge

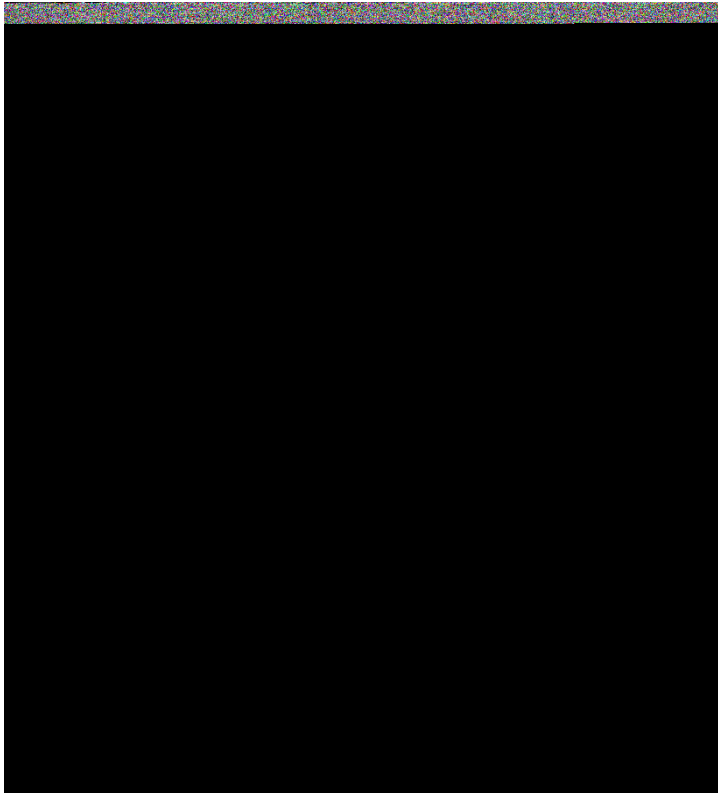


Plastic Hinge



Bending –Local Buckling

- Dominated by compression stresses

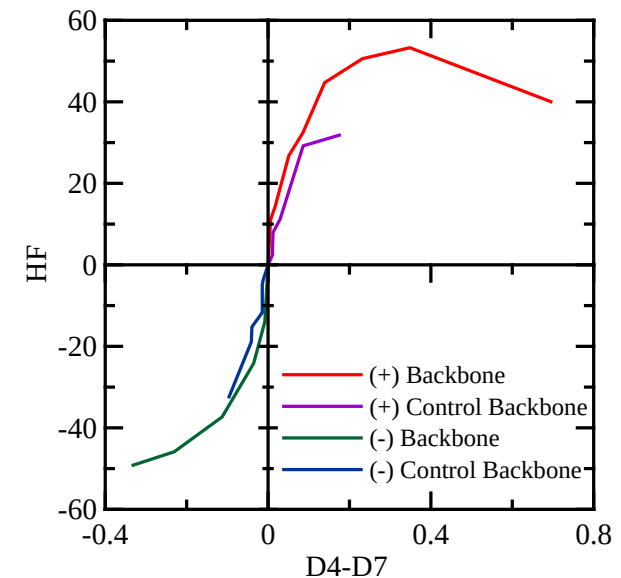


Large scale tests Test results, cont'd

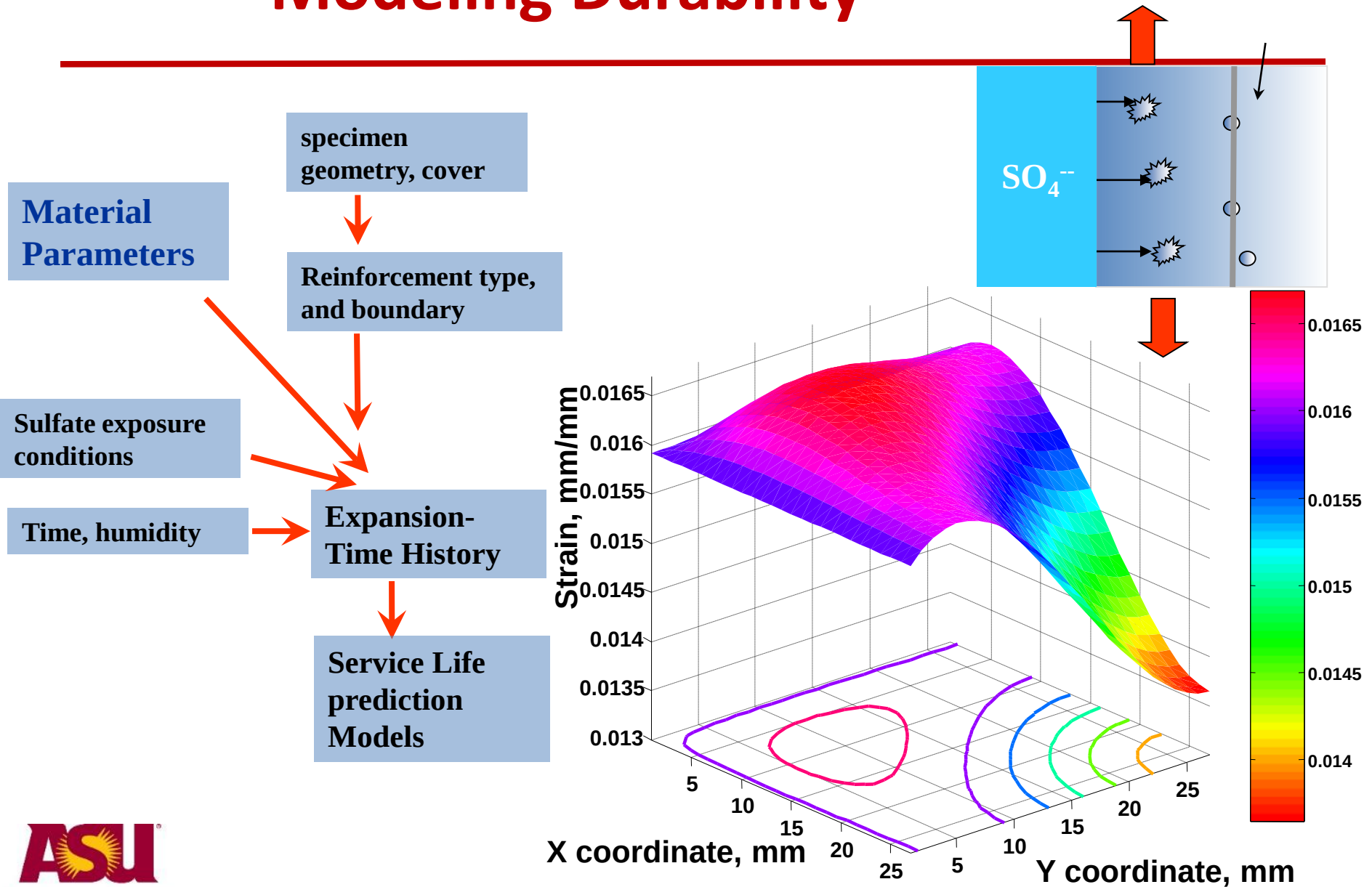
Backbone curves for strengthened wall vs. control



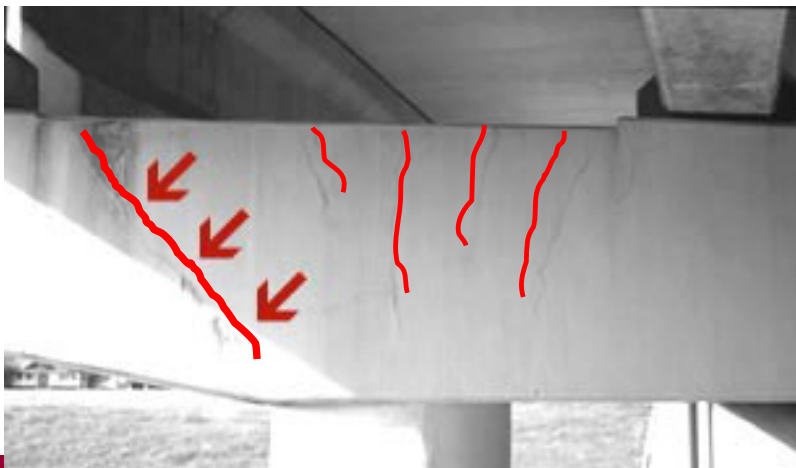
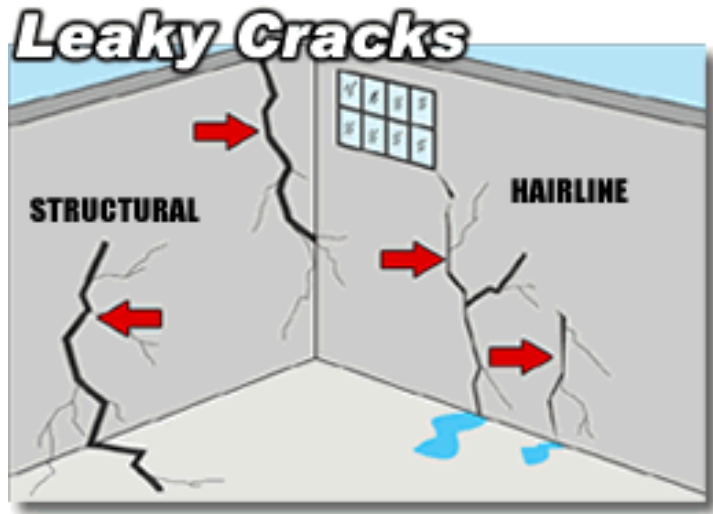
2 plies, 0/90



Modeling Durability



Types of cracks

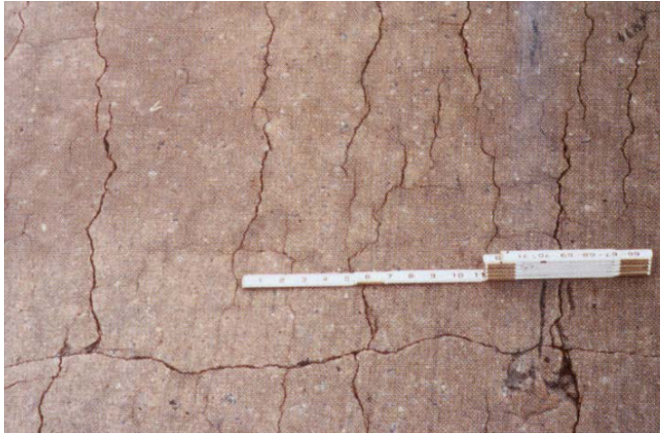


Structural related



Durability related

Concerns with Durability

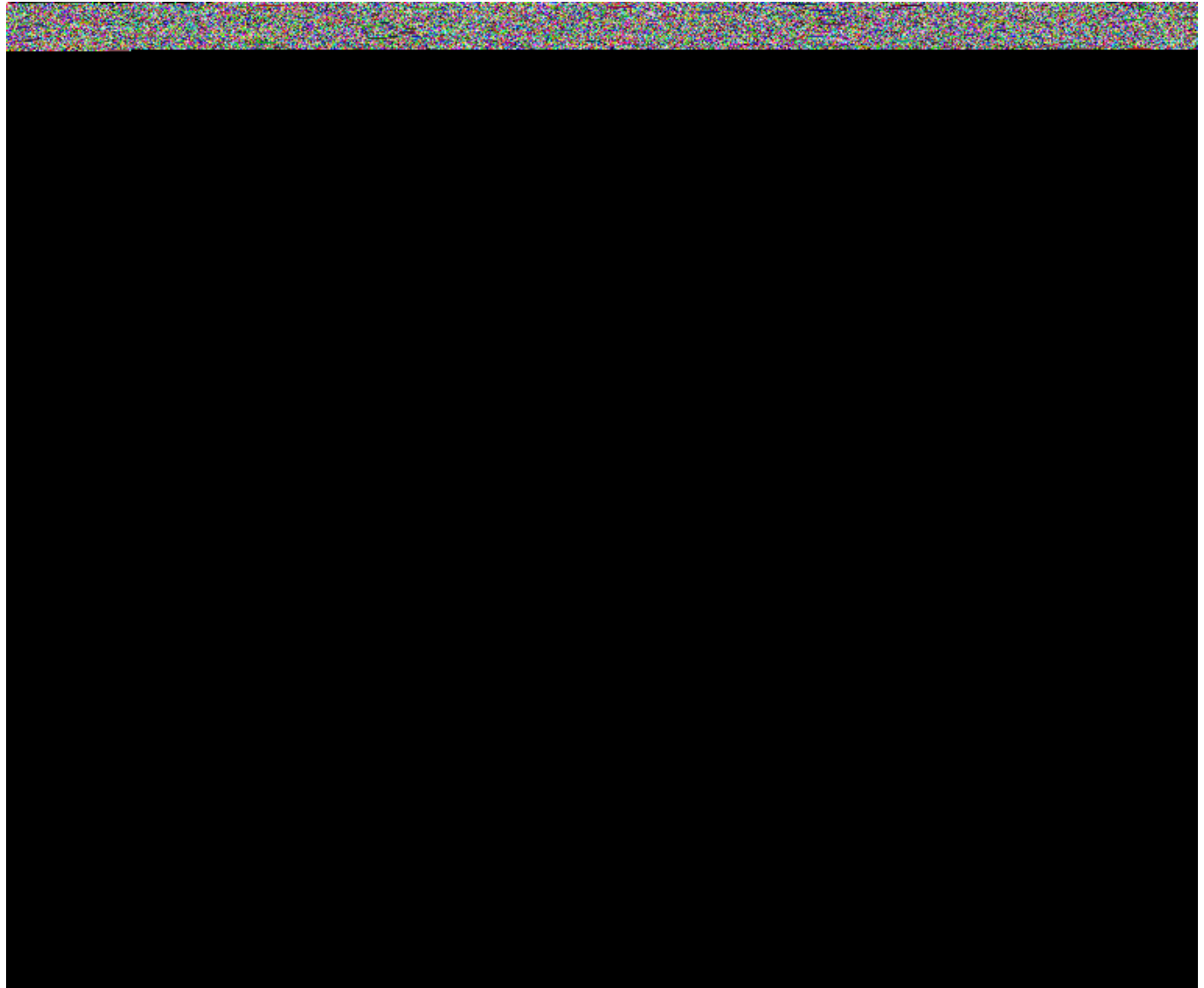


Bridge deck

Concerns with Structural integrity



Types of cracking



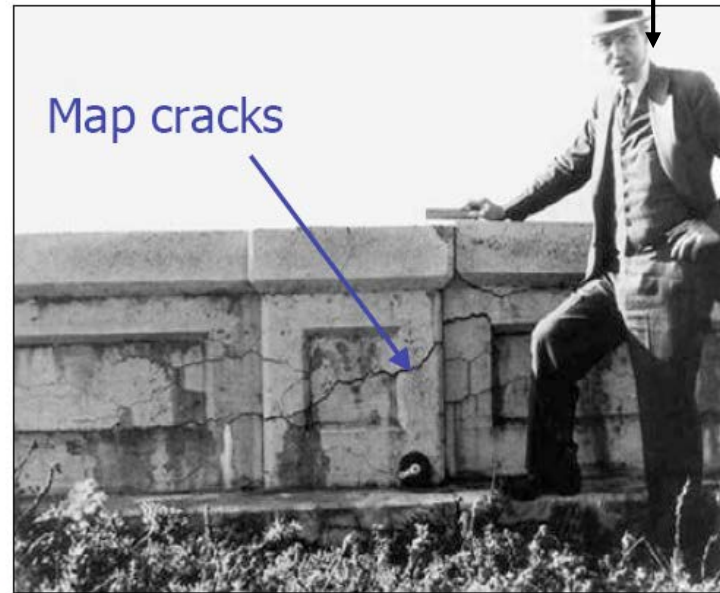
Alkali-Silica Reaction (ASR)



Vertical cracks

Damages in California Highway system

Thomas E. Stanton



Map cracks

California, 1936

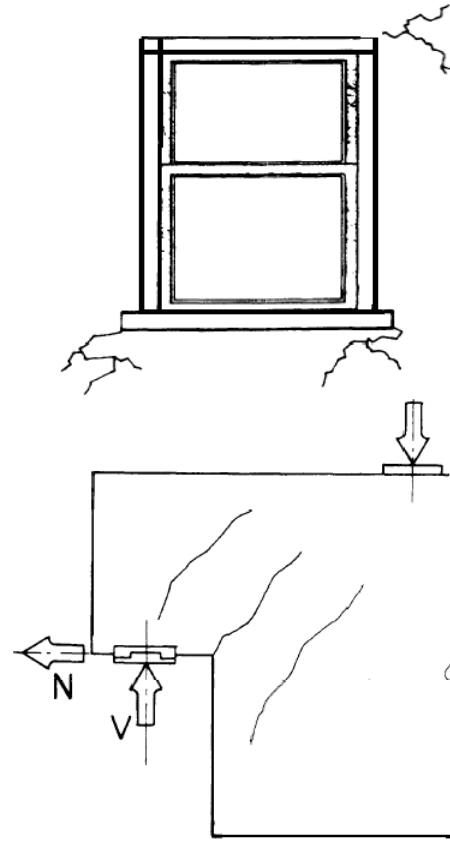
Mehta and Monteiro 2005

- **Cracks due to foundation settlement**
Diagonal cracks in concrete walls



Errors in design and detailing

- Re-entrant corners in walls, precast members and slabs
 - Need proper diagonal reinforcement
 - Common error- lightly reinforce because the wall may be non-structural
- Improper foundation design, resulting in differential movements



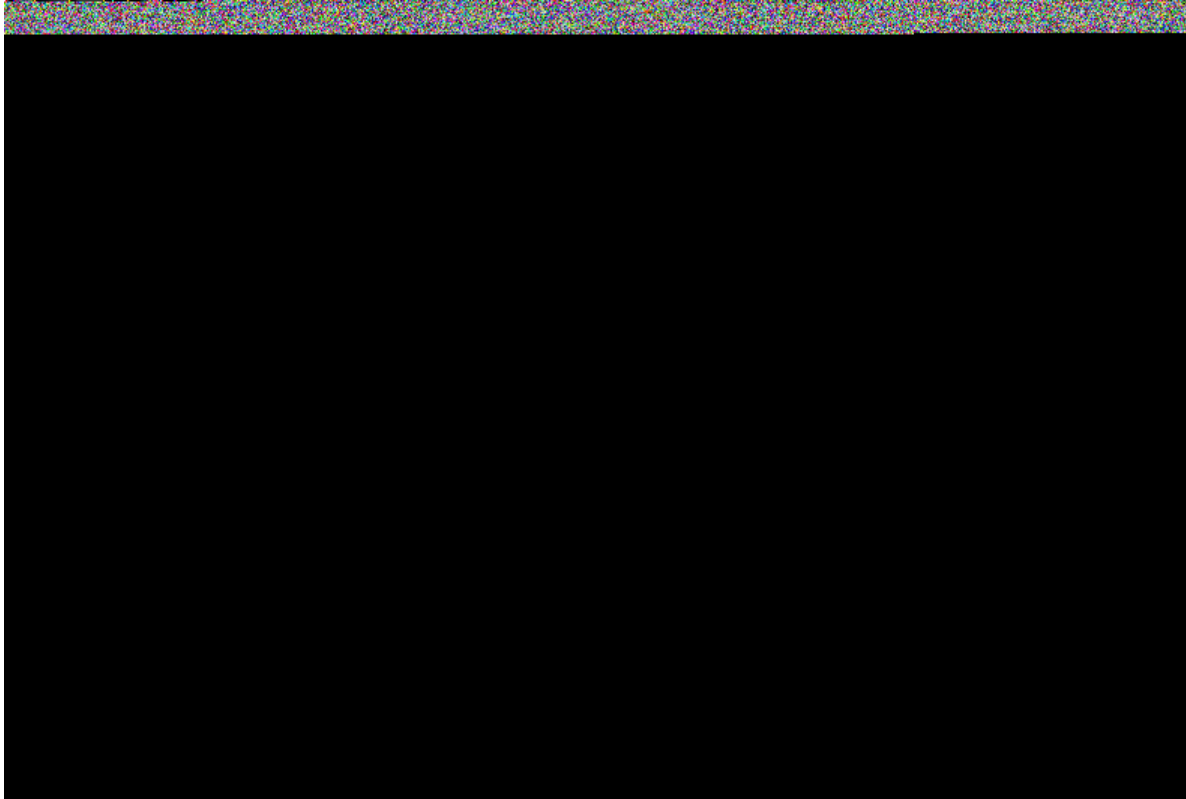
The Corrosion Problem



**Williamsburg Bridge - New York City,
NY; built in 1903**

- traveled by over 100,000 vehicles per day; train traffic on the deck below
- Inspected in 1991 and subsequently closed for 7 months
- direct cost to repair the corrosion problems: \$750 million
- indirect cost – to local businesses ; inconvenience and time of commuters estimated to be \$8.2 billion

Effects of Corrosion



Corrosion - Pop outs and Corrosion spalling



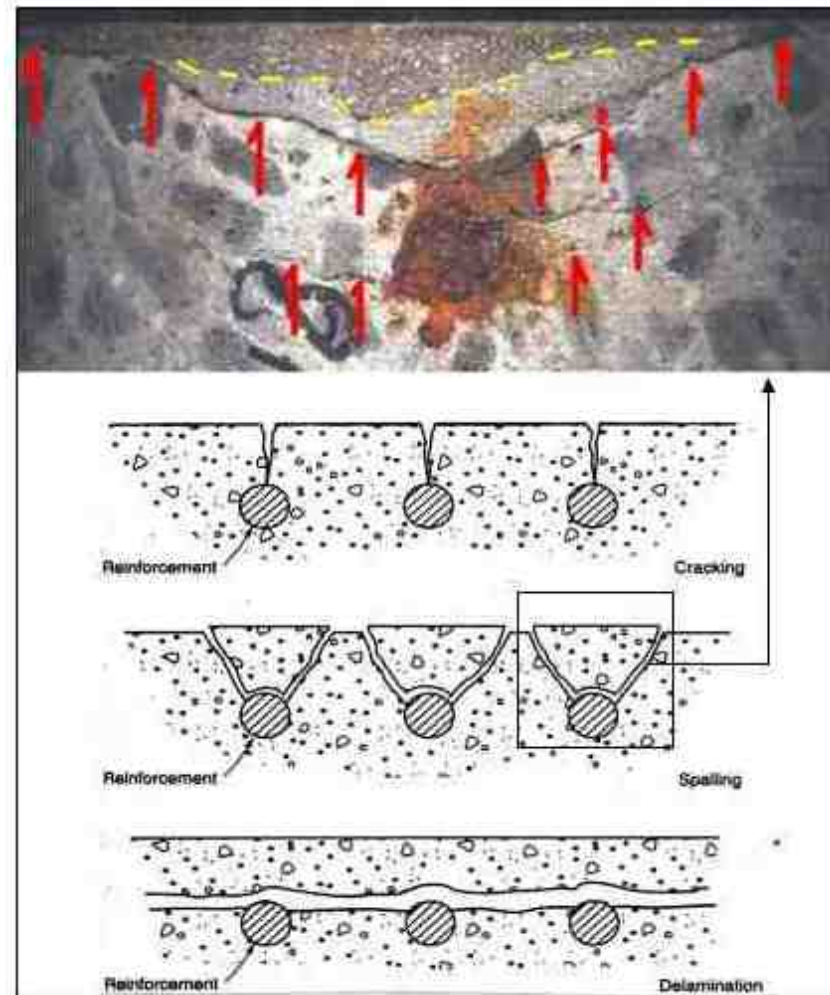
Atmospheric corrosion in Parking Structures

- Exposure to moisture, loss of coating
- Classic case of corrosion
- Pop outs and Corrosion spalling
- Sealing may not solve the problem once the corrosion has initiated



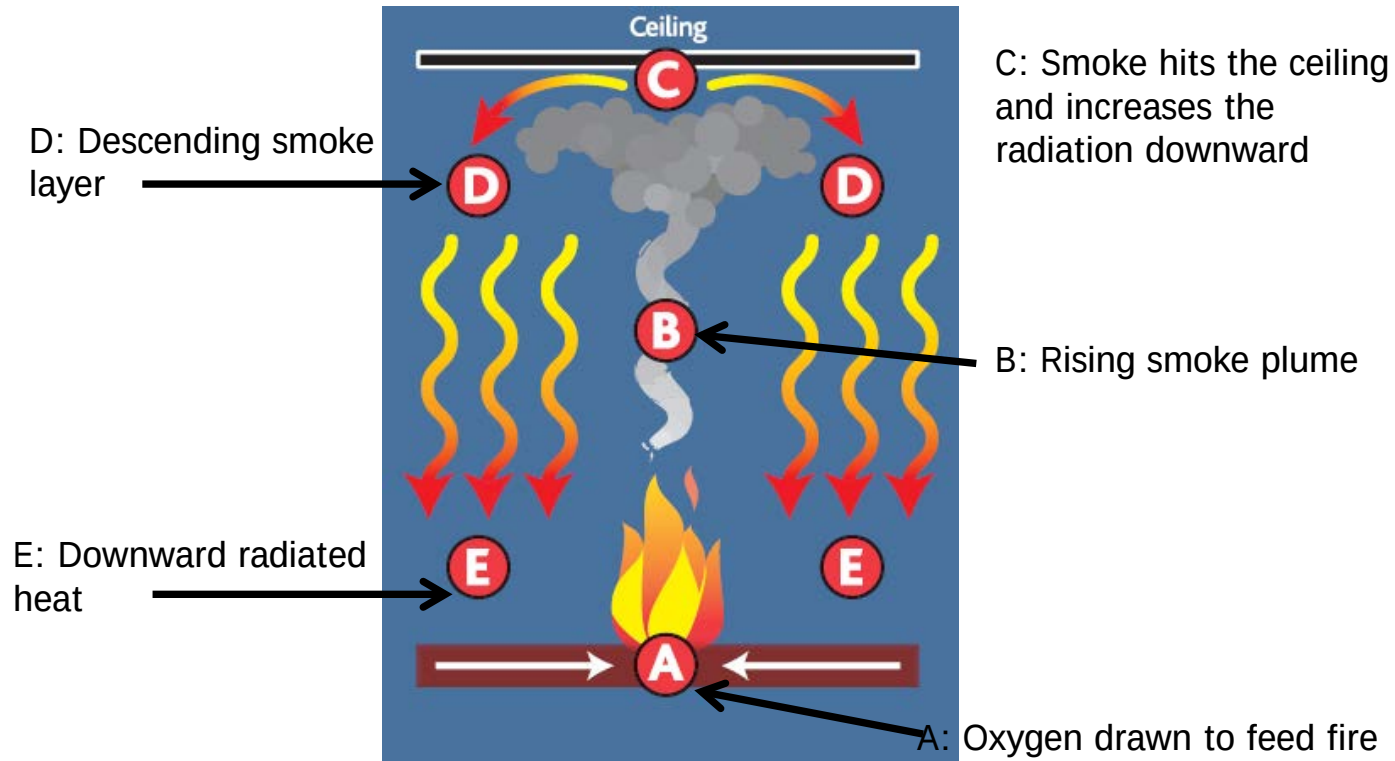
Corrosion due to Differential Environment

- Differences in chemical composition in the medium
- Different levels of oxygen content, salinities
- Usually just below the mean tide level
 - Steel piles
- Crevices and corners with less oxygen



Delamination due to corrosion

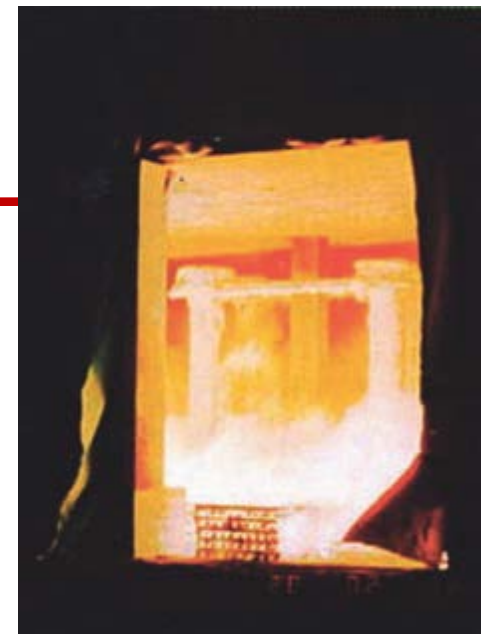
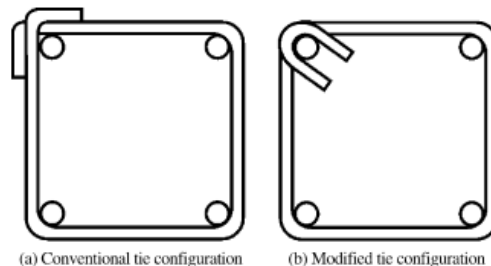
A Standard Compartment Fire



Concrete and Fire

Temperature (°C)	What happens
1000	
900	Air temperatures in fires rarely exceed this level, but flame temperatures can rise to 1200°C and beyond.
800	
700	
600	Above this temperature, concrete is not functioning at its full structural capacity.
550-600	Cement-based materials experience considerable creep and lose their loadbearing capacity.
400	
300	Strength loss starts, but in reality only the first few centimetres of concrete exposed to a fire will get any hotter than this, and internally the temperature is well below this.
250-420	Some spalling may take place, with pieces of concrete breaking away from the surface.

- In most cases, concrete does not require any additional protection because of its built-in resistance to fire
- It is non-combustible (i.e. it does not burn) and has a slow rate of heat transfer, which makes it a highly effective barrier to the spread of fire.



Conventional tie configuration

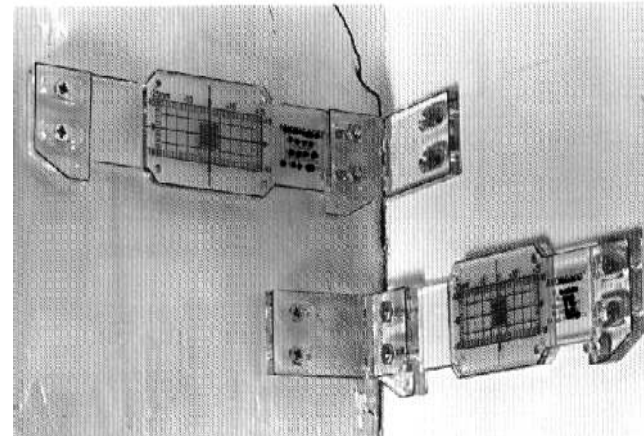
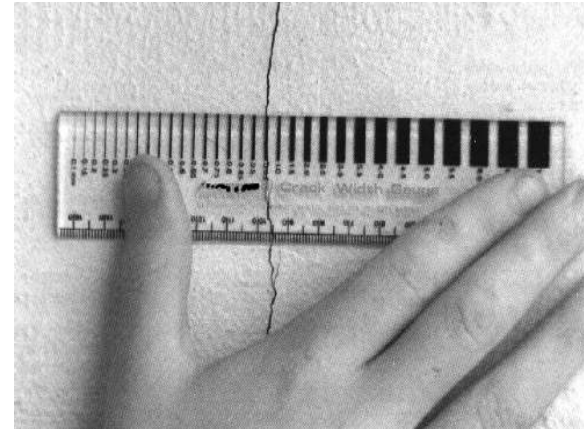


Modified tie configuration

Effect of confinement in fire resistance

Conventional Cracking Survey

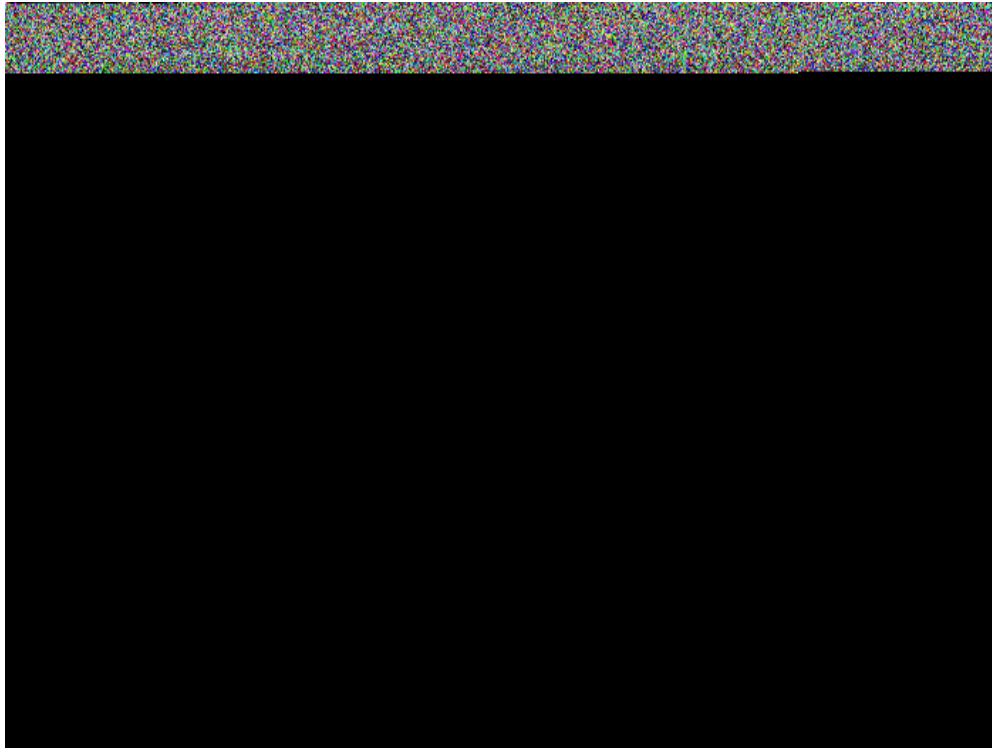
- Estimating surface crack length and width Crack
- Comparators, accuracy of 0.001 in
- Hand held microscope
- Initial and continued readings
- Depth and location of cracks

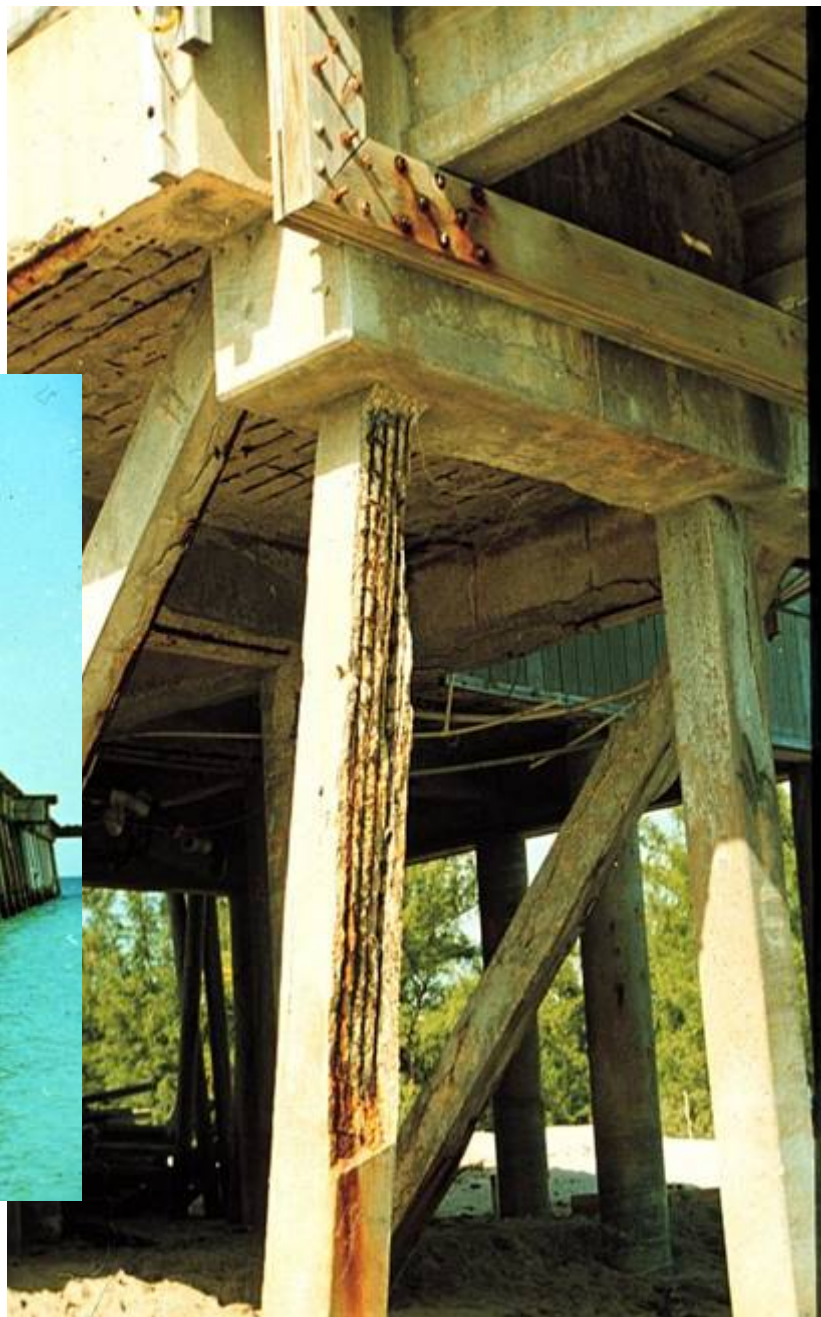


Corrosion and rehabilitation

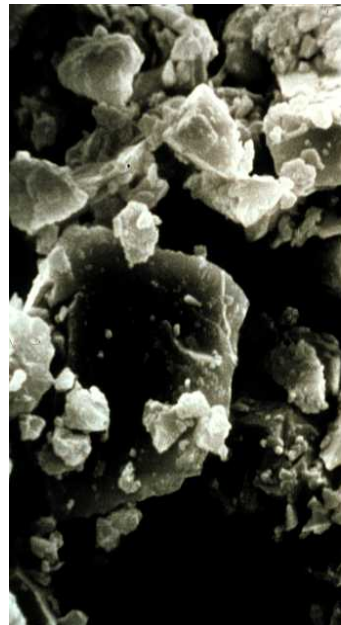
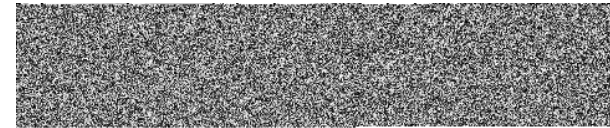
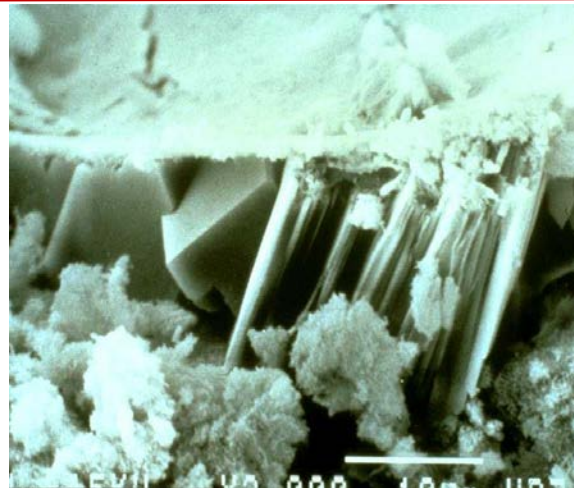
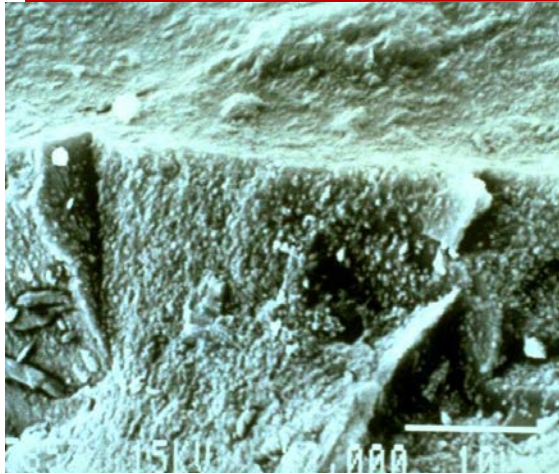


Durability Aspects





Reduce internal porosity, limit the ingress of harmful ions



Portland Cement

Silica Fume